Beyond the Missing Wedge: Uncertainty-Quantified 3D Lung Nodule Assessment from Low-Dose Digital Chest Tomosynthesis

Abstract

Lung nodules are followed up by measuring how fast they grow, but computed tomography (CT), the standard tool, gives the patient a radiation dose and a cost at every follow-up, while low-dose digital chest tomosynthesis (DTS) cannot resolve depth because its 30° sweep leaves a "missing wedge" of unmeasured information. MedLift-3D is a click-to-3D system that turns one click on a DTS slice into a 3D nodule measurement with a calibrated uncertainty. We first show from projection moments that every DTS view measures a nodule's mass exactly, while its depth extent, which volume needs, is about 15 times worse conditioned; mass and mass doubling time are therefore the measurements DTS can support. The pipeline simulates low-dose 15-view DTS for 583 LIDC-IDRI patients (1,144 radiologist-annotated nodules), reconstructs a 64 mm prism through the chest around the click, fills the missing wedge with a 17.4-million-parameter conditional 3D diffusion model kept consistent with the X-rays, represents the nodule with 3D Gaussians that give its mass in closed form, and wraps every number in conformal prediction intervals and calibrated size-category probabilities. A learned 3D detector proposes where to click and a seen/unseen (null-space) analysis elucidates on which part of every measurement originates from the X-rays and what is learned from the prior. The system components performed as expected: the PSNR of the nodule-region images on the held out data went from 11.88 dB to 15.23 dB after diffusion prior; the CPM of the detector was 0.298, and the baseline (training-free) was 0.017; the conformal intervals achieved 95% coverage on the 74 test nodules; and the calibrated category probabilities reduced the Brier score for a volume of at least 100 mm^3 from 0.54 to 0.17. The null space analysis validated that the X-ray measured nearly all of the mass of a nodule (0.3% unseen) and that the prior's unseen content is correlated with the truth (skill 0.22). However, the measurements did not achieve the desired target; mass was not more precise than volume, the volume error of the entire method (MdAPE 94.6%) was not smaller than the physics-only reconstruction (87.7 – 91.2%) and the valid ranges were not narrow enough for use. These failures we attribute to a prior that shrinks faint nodules, to the mass measurements on the image plane and not from the projections, and to the fact that measurements made in the calibration set were flawed, and demonstrate that better image quality did not necessarily lead to better measurement.

Keywords: digital tomosynthesis, lung nodule volumetry, diffusion models, 3D Gaussians, conformal prediction, explainable AI

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Chapter 1

Introduction

This chapter states the clinical problem that MedLift-3D solves and why it matters. It then lists the objectives as testable hypotheses, summarises the method and its outcomes, and outlines the rest of the report.

1.1 Project Overview

Lung cancer is the leading cause of cancer death worldwide, with about 2.5 million new cases (12.4% of all cancers) and 1.8 million deaths (18.7% of cancer deaths) in 2022 [1]. Survival depends strongly on the stage at diagnosis: five-year survival is 70–90% for stage I non-small cell lung cancer but only 15–19% for stage IV [2]. Early lung cancer usually appears as a pulmonary nodule, yet most nodules are benign. Radiologists therefore decide by following a nodule over months and measuring how fast it grows, usually as a volume doubling time (VDT) [3, 4].

Computed tomography (CT) is the standard tool for this follow-up, but every scan gives the patient a radiation dose of several millisieverts and costs money and scanner time [5]. Low-dose CT lowers the dose, yet repeated follow-up still adds up: about four radiation-induced cancers per 10,000 screened people, most of it from the follow-up scans [6]. In low-resource settings such as rural Bangladesh, CT is often not available at all.

Digital chest tomosynthesis (DTS) acquires 15–74 low-dose X-ray views during a short tube sweep of about 30°, at roughly 0.1 mSv, on equipment that costs far less than CT [7]. It finds nodules about three times more often than chest radiography [8]. However, a 30° sweep leaves a *missing wedge* of directions that no X-ray passes through (Fig. 1.1), so DTS cannot resolve depth and every 3D size measured from it is partly a guess.

Problem statement. Given a low-dose chest DTS acquisition and one click on a nodule, produce its 3D mass, volume and growth rate together with an uncertainty that is honest, so that a clinician can follow nodules without repeated CT and knows when a measurement cannot be trusted.

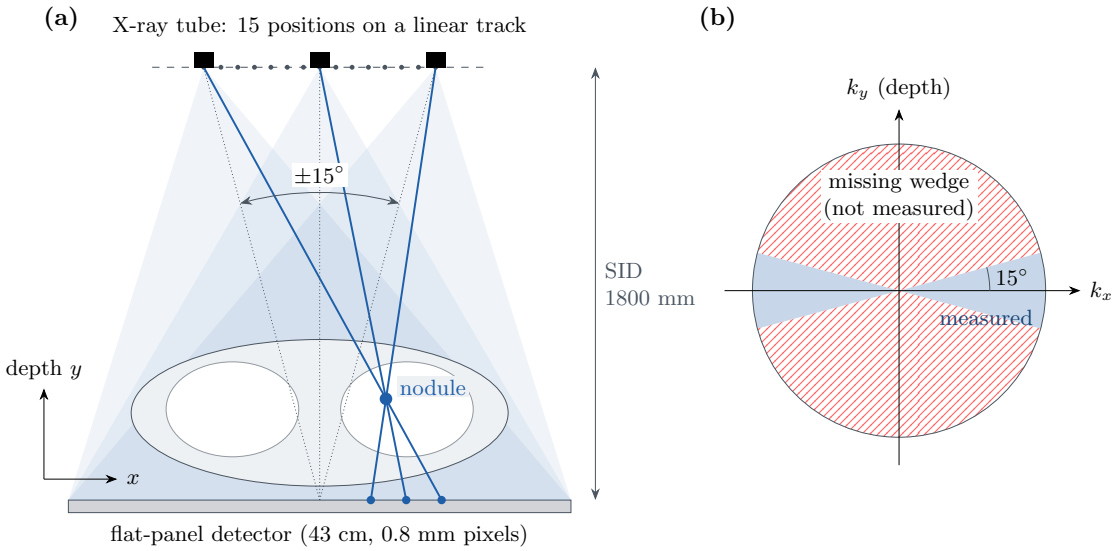


Figure 1.1: Chest DTS acquisition. (a) The X-ray tube moves along a linear track over ±15° and the nodule's shadow shifts on the detector with the tube angle; this shift is the only depth information. (b) In Fourier space each view measures one plane, so a ±15° sweep measures only a double wedge around the in-plane axis and leaves the depth frequencies unmeasured.

1.2 Motivation

This project is motivated by four observations: nodules need to be followed over time, the size of nodules is uncertain when measured in 2D, follow-up using CT is costly because of radiation, access, and existing reconstruction methods are not able to produce trustworthy numbers from DTS. Our approach is guided by a 5th, computational observation.

1.2.1 Clinical Importance of Nodule Follow-up

The 5-year survival rates are 64% for localised lung cancer, and 9% for distant disease [9, 10]. The key distinguishing feature between malignant and benign nodules is growth and the guidelines are based on ranges of volume and doubling times: $<$ 100, 100–300 and $\geq$300 mm^3; $<$ 400, 400–600 and $>$ 600 days [4, 11].

1.2.2 Limits of 2D Measurement

Diameters measured on single slices assume a sphere. Volumes estimated from 2D measurements differ from 3D volumetry by 47–85% on average [12], 2D measurements of small nodules are unreliable [13], and a 1 mm change in diameter is a 33% change in volume [14]. Measurement error larger than the true growth delays treatment.

1.2.3 Radiation and Access

CT follow-up means repeated dose and requires well-equipped centres [5, 15]. In Bangladesh, CT scanners are concentrated in divisional cities and large hospitals, so many patients travel long distances and pay for every follow-up scan, while digital

radiography rooms, which can be upgraded to tomosynthesis, are far more widespread. DTS is cheaper, faster and much lower in dose than CT, but today it offers no trustworthy 3D measurement, so a nodule found on DTS is still sent to CT for every follow-up.

1.2.4 Why Existing Reconstruction and AI Are Not Enough

Iterative reconstruction from a 30° sweep results in 3–5 times nodule extension along the depth axis in comparison to in-plane (the median was 3.5 in our measurements of LIDC nodules). The missing wedge can be filled by generative models, but a cancer-following model that creates anatomy is hazardous [16, 17]. Sparse view methods have been published with image quality (PSNR,SSIM) values not calibrated clinical numbers and none is developed for DTS geometry.

1.2.5 Computational Motivation

Not only what it can show, but we asked what DTS can *measure*. By derivation of projection moments (Section 3.2) it is shown that every DTS view measures the total excess attenuation of a nodule exactly, its depth extent, which volume needs, is about 15 times worse conditioned for a ±15° sweep. A system for DTS should therefore start with a statement of *mass*, then a fill depth with a learned prior that is consistent with the X-rays, and a statement of uncertainty for each of the numbers. The use of mass growth as a CT biomarker is already proven to detect growth earlier than volume [18].

1.3 Objectives

The objectives are stated as testable hypotheses (H1–H6); any of them could turn out false, and Chapter 4 reports what we found.

O1 Mass-first measurement (H1). Show, in theory and on data, that nodule mass from DTS has a smaller error than nodule volume.

O2 Learned prior for the missing wedge (H2). Train a conditional 3D diffusion model that, combined with physics data consistency and 3D Gaussians, reduces volume error compared with physics-only reconstruction.

O3 Calibrated uncertainty (H3, H5). Give every mass and volume a conformal 90% prediction interval that reaches its nominal coverage on held-out patients, and give calibrated probabilities of the clinical size categories with coverage in every size group.

O4 Growth (H4). Measure growth on virtual follow-up scans and test whether mass doubling time (MDT) classifies growth more accurately than VDT.

O5 Explainability (H6). Show for each nodule which part of the 3D result was measured by the X-rays (seen) and which was supplied by the prior (unseen), and test

whether the prior's unseen content recovers the truth rather than inventing it.

O6 Usable system. Deliver a click-to-3D application with a learned detector that suggests where to click, running on free or low-cost cloud hardware.

1.4 Methodology

MedLift-3D is evaluated as a virtual clinical trial on real patient CT, because no public dataset has chest DTS together with CT ground truth. Figure 3.6 in Chapter 3 shows the full pipeline.

1. **Data:** 583 LIDC-IDRI patients with 1,144 nodules of 4–30 mm marked by at least three of four radiologists [19, 20], split by patient 60/20/20 into training, calibration and test sets.
2. **DTS simulation:** 15 low-dose views over $\pm 15^\circ$ with Poisson noise, reconstructed by SIRT [21].
3. **Click and prism:** the clinician, helped by a learned detector, selects a nodule; a 64 mm column through the whole chest (the *prism*) is reconstructed around it.
4. **Diffusion prior:** a 17.4-million-parameter conditional 3D U-Net diffusion model [22–24] generates eight CT-like hypotheses of the prism, each kept consistent with the X-rays by SIRT steps inside the sampler.
5. **3D Gaussians:** each hypothesis is fitted with anisotropic 3D Gaussians [25, 26], which give the nodule's mass in closed form.
6. **Uncertainty:** the spread of the eight hypotheses is calibrated into conformal intervals and predictive distributions on the calibration patients [27–30].
7. **Explanation:** each result is split into seen and unseen parts by a conjugate-gradient null-space decomposition [16, 31], and the detector is explained with input×gradient saliency [32].

1.5 Project Outcome

The project generates the following three sets of outputs: a technical system that is shown to operate; a system usable by a clinician; and a research contribution.

1.5.1 Technical Outcome

An automated, step-by-step, scripted process from data download to final tables, figures and HTML report, which has been verified and passed. It includes a differentiable DTS simulator and projector, whole-chest and prism reconstruction, a trained conditional diffusion prior and nodule detector, Gaussian measurement, conformal calibration and an

explainability module, with 69 unit tests documented and with free use of cloud GPUs via a rental service and Kaggle notebooks.

1.5.2 Clinical Outcome

A click-to-3D application that returns, in under a minute, the nodule's mass, volume, long-axis diameter and EUPS size category with calibrated probabilities, together with a virtual follow-up for growth. Every number comes with an interval instead of a single value, and a low-confidence flag tells the clinician when the DTS evidence is too weak and a CT is advisable. The tool is meant for following known nodules, the step that today requires repeated CT.

1.5.3 Research Contribution

(i) A projection-moment argument that DTS measures mass but not depth extent; (ii) a data-consistent diffusion prior inside a DTS prism combined with Gaussian mass; (iii) calibrated, decision-level uncertainty for DTS volumetry; (iv) a truth-referenced seen/unseen test of what the prior adds. Section 2.3 states exactly what is and is not new.

1.6 Organization of the Report

Chapter 2 provides the background of DTS, diffusion models, 3D Gaussians, conformal prediction and explainability, discusses related applications and research, and identifies the gap. The requirements, context and data flow diagrams, user interface, detailed method, including alternatives examined, project plan and task allocation are presented in Chapter 3. Chapter 4 describes the environment and tools as well as how each requirement was verified and the results of H1-H6. Chapter 5 discusses standards, issues related to society, health and safety, law, culture, ethics, environment and sustainability, costs and the complex engineering problem mapping. In Chapter 6 the work is summarised, limitations are pointed out, future work is suggested and what has been learnt is summarised.

Chapter 2

Background

This chapter gives the background needed for the rest of the report and reviews existing applications and research. It concludes in the gap that MedLift-3D fills.

2.1 Preliminaries

This section presents the physics of X-ray projection and tomosynthesis, iterative reconstruction and null space, diffusion models, Gaussian representations and nodule mass, conformal prediction, and explainability and detection measures which are used throughout the remainder of the report.

2.1.1 X-ray Projection and Digital Tomosynthesis

A detector pixel records the line integral of the linear attenuation coefficient μ along the ray from the source. With I_0 incident photons the expected count is

$$\bar{I} = I_0 \exp\Big(- \int_{\text{ray}} \mu(\mathbf{r})\, \mathrm{d}l \Big), \qquad I \sim \text{Poisson}(\bar{I}), \tag{2.1}$$

so fewer photons (lower dose) mean more noise. CT values are expressed in Hounsfield units, $\text{HU} = 1000\,(\mu - \mu_{\text{water}})/\mu_{\text{water}}$.

The missing wedge. CT collects projections over 180° or more. Chest DTS moves the tube along a short linear track, typically ±15° (GE VolumeRAD) to ±20° (Shimadzu SonialVision), with 15–74 views [7]. By the Fourier slice theorem each view measures one plane of the object's 3D Fourier transform, so a narrow sweep leaves a wedge of depth frequencies unmeasured (Fig. 1.1b). Reconstructions are sharp in the coronal plane but every structure is smeared in depth.

2.1.2 Iterative Reconstruction and the Null Space

Writing the projection operator as A, the measured line integrals as $\mathbf{b}$ and the volume as $\mathbf{x}$, SIRT [21, 33] solves $A\mathbf{x} \approx \mathbf{b}$ by repeated normalised back-projection of the residual:

$$\mathbf{x}^{(k+1)} = \mathbf{x}^{(k)} + \lambda C A^{\top} R\big(\mathbf{b} - A\mathbf{x}^{(k)}\big), \tag{2.2}$$

where R and C hold the inverse row and column sums of A. It is stable under noise but cannot recover anything that lies in the null space of A.

Seen and unseen parts. Any volume splits uniquely as $\mathbf{x} = \mathbf{x}_{\text{seen}} + \mathbf{x}_{\text{unseen}}$ with $\mathbf{x}_{\text{seen}} \in \mathcal{R}(A^{\top})$ and $\mathbf{x}_{\text{unseen}} \in \mathcal{N}(A)$, i.e. $A\mathbf{x}_{\text{unseen}} = 0$. The X-rays determine only the seen part; the unseen part of any reconstruction comes from its prior [16, 31]. Figure 2.1 illustrates this.

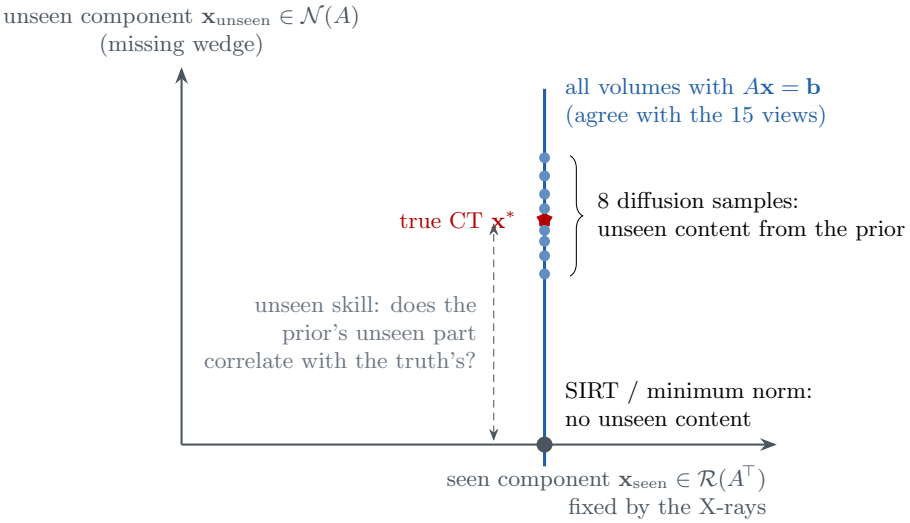


Figure 2.1: Seen (range) and unseen (null-space) components. All volumes on the vertical line agree with the measured views. SIRT returns the minimum-norm point with no unseen content; diffusion samples supply unseen content, which we test against the true CT (unseen skill, Section 3.2).

2.1.3 Denoising Diffusion Models

A diffusion model learns to reverse a gradual noising process [22]. With a cosine noise schedule $\bar{\alpha}_t$ [34], a noisy sample is

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\,\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\,\boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, I). \tag{2.3}$$

We train the network to predict the velocity $\mathbf{v} = \sqrt{\bar{\alpha}_t}\,\boldsymbol{\epsilon} - \sqrt{1-\bar{\alpha}_t}\,\mathbf{x}_0$ (v-prediction [24]), minimising

$$\mathcal{L}(\theta) = \mathbb{E}_{\mathbf{x}_0,\mathbf{c},t,\boldsymbol{\epsilon}} \left\| \mathbf{v} - \hat{\mathbf{v}}_\theta(\mathbf{x}_t, \mathbf{c}, t) \right\|_2^2, \tag{2.4}$$

where the condition $\mathbf{c}$ is the DTS reconstruction, so the model learns $p(\mathbf{x}_0 \mid \mathbf{c})$. DDIM [23] samples in tens of steps instead of a thousand, and data-consistency (DC) sampling corrects every denoised estimate with the physics so that samples agree with the measurements [35, 36].

2.1.4 3D Gaussian Representations and Nodule Mass

A volume can be written as a sum of anisotropic Gaussians with centres $\boldsymbol{\mu}_i$, covariances $\Sigma_i = R_i S_i S_i^\top R_i^\top$ and amplitudes a_i [25, 26]:

$$f(\mathbf{r}) = \sum_i a_i \exp\Big(-\tfrac{1}{2}(\mathbf{r}-\boldsymbol{\mu}_i)^\top \Sigma_i^{-1}(\mathbf{r}-\boldsymbol{\mu}_i)\Big), \qquad M = \int f = \sum_i a_i (2\pi)^{3/2} s_{i1} s_{i2} s_{i3}, \quad (2.5)$$

where s_{i1}, s_{i2}, s_{i3} are the Gaussian's standard deviations. The mass is therefore closed form, and the field can be evaluated on any grid.

Nodule mass. In this report mass is the nodule's excess attenuation over the surrounding lung, converted to milligrams with the CT density scale (about 0.7–0.85 mg/mm^3 on LIDC nodules). Mass doubling time is computed like VDT but from mass; it detects growth of part-solid nodules earlier [18].

2.1.5 Conformal Prediction

Split conformal prediction [27, 30] turns any predictor into intervals with a finite-sample guarantee: if calibration and test cases are exchangeable, a $(1-\alpha)$ interval contains the truth with probability at least $1-\alpha$. A *normalised* version scales each interval by a difficulty estimate (here the spread of the diffusion hypotheses); a *Mondrian* version calibrates within groups (here nodule size groups) so that coverage holds in every group; and *conformal predictive systems* (CPS) return a full calibrated cumulative distribution for each case [28, 29], from which calibrated probabilities such as $P(V \geq 300\ \text{mm}^3)$ follow. Figure 2.2 shows the procedure.

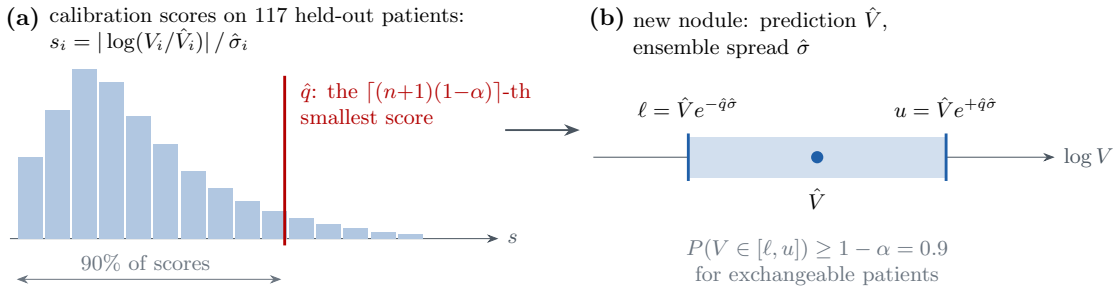


Figure 2.2: Split-conformal calibration (schematic). (a) Normalised scores on held-out calibration patients and their $(1-\alpha)$ quantile $\hat{q}$. (b) The interval for a new nodule, valid for exchangeable patients without any assumption on the error distribution.

2.1.6 Explainability and Detection Metrics

Input×gradient saliency [32] multiplies each input voxel by the gradient of the output with respect to it, showing which voxels drove a detector's score. Detection is measured by the Competition Performance Metric (CPM), the mean sensitivity at 1/8, 1/4, 1/2, 1, 2, 4 and 8 false positives per scan [37].

2.2 Literature Review

The review has two parts: applications that already address parts of the problem in practice, and research on radiation and screening, nodule measurement, chest DTS, deep priors for sparse-view reconstruction, Gaussian tomography, and trustworthiness and uncertainty.

2.2.1 Similar Applications

Parts of the problem are already available in clinical practice, but these tools do not measure nodules in 3D from DTS. The GE and Shimadzu chest DTS units generate coronal slices which radiologists read visually, and nodule size is either estimated visually or a diameter is provided in a single slice. CT nodule software, offered by scanner vendors and AI companies, segments nodules and report volumes and doubling time – but only on CT. Detection systems like the ones compared in the LUNA16 challenge predict the locations of nodules but do not measure them. Manual segmentation is possible with general tools, like 3D Slicer, ITK-SNAP, etc., which do not carry out any learnt prior or uncertainty, while the research codes for sparse-view reconstruction generate the whole volumes for CT geometry and report only the image quality. These applications are summarised in Table 2.1 and each has a drawback for our problem.

Table 2.1: Applications similar to MedLift-3D.

Application	What it does	What it lacks for our problem
Chest DTS units (GE VolumeRAD, Shimadzu SonialVision)	Acquire and reconstruct DTS slices for reading	No 3D nodule measurement; depth smear left to the reader
CT nodule volumetry software (vendor lung packages, AI nodule tools)	Detect and segment nodules on CT, report volume and VDT	Requires CT at every follow-up; no DTS support
CT computer-aided detection (LUNA16-style) [37]	Suggest nodule locations on CT	Detection only; no measurement or uncertainty
3D Slicer, ITK-SNAP	Viewing and manual segmentation	No learned prior, no calibrated numbers
Research sparse-view reconstruction codes (R^2-Gaussian, 3DGR-CT) [26, 38]	Reconstruct whole volumes from few views	CT geometry, image metrics only, no clinical numbers

2.2.2 Related Research

Radiation and screening. Mascalchi and Sali [15] reviewed the cumulative dose of low-dose CT screening (9.3–13.0 mSv over ten years) and argued for ALARA and ultra-low-dose methods; Rampinelli et al. [6] estimated the cancer risk from screening and follow-up CT.

3D measurement of nodules. Lee [14] and de Margerie-Mellon et al. [12] showed that volumetry is more reproducible than diameters but varies between software, and Revel et al. [13] showed that 2D measurements of small nodules are unreliable. NELSON [4] and the European position statement [11] define the volume and VDT categories we use. De Hoop et al. [18] showed that mass detects growth of ground-glass nodules earlier than volume or diameter.

Chest DTS. Vikgren et al. [8] found that DTS detected about three times more nodules than chest radiography, and Dobbins and Godfrey [7] described the depth-resolution limit set by the sweep angle. DTS studies measure nodules in 2D or by reader judgement.

Sparse-view CT with deep priors. Score-based and diffusion models have been applied in the sinogram domain [39], with wavelet decompositions [40, 41], for spectral CT [42] and zero-shot with variational solvers [43]; DPS and DDS add data consistency to sampling [35, 36], and foundation models learn lung CT priors from 100,000 scans [44]. Classical iterative reconstruction [33] and compressed sensing [45] suppress noise but over-smooth fine detail, and implicit neural fields [46] are slow per patient. Lighter physics-conditioned diffusion [47], deep image prior with ADMM [48], mixed priors for metal

artifacts [49], learned view sampling [50] and unsupervised low-dose denoising [51] address other parts of the problem. All of these report image quality on CT volumes; none produces a calibrated clinical measurement or uses DTS geometry.

3D Gaussians in tomography. 3DGR-CT [38] and R^2-Gaussian [26] reconstruct sparse-view CT with Gaussians, and spatiotemporal Gaussians handle motion [52, 53]. They use Gaussians for speed and quality, not for closed-form mass, and have no learned prior for the missing wedge.

Trustworthiness and uncertainty. Bhadra et al. [16] defined hallucinations through the null space of the imaging operator, DDNM [31] uses a range–null-space decomposition inside diffusion sampling, and Zhang et al. [17] used uncertainty estimates to suppress artifacts in limited-data CT. Conformal predictive systems are available in the open-source crepes library [29] but have not been applied to imaging biomarkers from DTS.

Table 2.2: Summary of the most related research.

Work	Method	Key finding	Gap for DTS nodule measurement
Li et al. [38]	FBP-guided 3D Gaussians	Faster than neural fields	Whole-volume CT; no prior; no measurement
Zha et al. [26]	Radiative Gaussian splatting	Accurate volume integration	No learned prior; no uncertainty
Guan et al. [39]	Sinogram-domain diffusion	Handles varying sparsity	2D; no 3D consistency
He et al. [43]	Zero-shot variational score solver	No paired data needed	Heavy compute; image metrics only
Chung et al. [35, 36]	DPS / DDS sampling	Measurement-consistent samples	Not for DTS; not calibrated
Wang and Guo [44]	Lung CT foundation model	Strong transfer	CT only
Bhadra et al. [16]	Null-space hallucination maps	Formal definition of hallucination	No truth-referenced skill per nodule
Zhang et al. [17]	Progressive artifact learning	Fewer false structures	Uncertainty not calibrated
de Hoop et al. [18]	Nodule mass on CT	Mass detects growth earlier	CT only
Boström [29]	Conformal predictive systems	Valid CDFs for any regressor	Not applied to imaging biomarkers

2.3 Gap Analysis

The review shows four gaps. First, DTS research stops at 2D detection and size; no work argues from the projection physics which 3D quantity DTS can measure. Second, generative sparse-view methods target whole CT volumes and image metrics, not a clinical number for one nodule, and none is built for the DTS sweep. Third, Gaussian tomography uses Gaussians for speed, not for closed-form mass. Fourth, reconstruction uncertainty is usually raw variance, which is not calibrated, and explanations of what a prior invented are not checked against the truth. Table 2.3 compares the closest works.

Table 2.3: Domain comparison: ✓ addressed, ✗ not addressed, ○ partially.

Work	DTS geometry	Learned prior	Data consistency	Gaussian mass	Clinical biomarker	Calibrated uncertainty	Seen/unseen explanation
Li et al. [38]	✗	✗	✓	✗	✗	✗	✗
Zha et al. [26]	✗	✗	✓	✗	✗	✗	✗
He et al. [43]	✗	✓	✓	✗	✗	✗	✗
Chung et al. [35]	✗	✓	✓	✗	✗	✗	✗
Zhang et al. [17]	✗	✓	✓	✗	✗	○	✗
Bhadra et al. [16]	✗	✗	–	✗	✗	✗	✓
de Hoop et al. [18]	✗	✗	–	✗	✓	✗	✗
MedLift-3D	✓	✓	✓	✓	✓	✓	✓

What we do not claim. 3D Gaussians in medical imaging, uncertainty for Gaussian reconstructions, conformal prediction in CT and for lung nodules, null-space hallucination analysis and saliency maps all exist already. Our contribution is their combination for DTS nodule measurement, the mass-first argument, and a calibrated, truth-checked evaluation.

2.4 Summary

DTS is low-dose and affordable but blind in depth, and existing AI reconstruction neither respects which quantities DTS can measure nor states how far its numbers can be trusted. MedLift-3D addresses this gap by measuring mass first, filling depth with a data-consistent diffusion prior, calibrating every number with conformal methods, and explaining what was measured and what was supplied by the prior. Chapter 3 turns this into requirements and a design.

Chapter 3

Project Design

This chapter turns the problem into requirements, models the system with context, data flow and decomposition diagrams, and explains the method together with the alternatives we rejected. It ends with the project plan and task allocation.

3.1 Requirement Analysis

Requirements elicitation. Requirements were gathered from four sources: (i) the clinical guidelines that radiologists follow for nodules, namely the EUPS volume categories and the NELSON doubling-time categories [4, 11]; (ii) meetings with our supervisor; (iii) the published limits of DTS and sparse-view reconstruction (Chapter 2); and (iv) the constraints of a student project: public data only, no clinical scanner, limited computing resources and a fixed deadline. Table 3.1 lists the stakeholders and their needs.

Table 3.1: Stakeholders and their needs.

Stakeholder	Need	How MedLift-3D answers it
Radiologist	A 3D size and growth rate from a low-dose exam, and to know when not to trust it	Mass, volume, category and doubling time, each with a calibrated interval and a low-confidence flag
Patient	Less radiation and cost per follow-up	DTS at about 0.1 mSv instead of CT
Low-resource hospital	Affordable equipment and software	DTS hardware; software runs on a single free or rented GPU
Examiner / researcher	Claims that can be checked	Every step ends in PASS/FAIL checks; test patients never used for tuning

Requirements analysis and negotiation. Several requirements conflict; Table 3.2 records how each conflict was resolved.

Table 3.2: Conflicting requirements and their resolution.

Conflict	Decision	Rationale
Low dose vs accuracy	Simulate $I_0 = 2 \times 10^4$ photons per pixel and view (about 1/20–1/40 of clinical DTS dose)	Conservative: a method that works at this dose works at clinical dose
Clinical view count (60–74) vs compute	15 views at equal total dose	Measured: the view count barely changes depth smear at equal dose (5.13 vs 5.34)
Fill the missing wedge vs never invent anatomy	Diffusion prior with X-ray data consistency, plus seen/unseen explanation	Every hypothesis agrees with the X-rays; what the prior adds is shown and tested
Single number vs honest uncertainty	Eight hypotheses + conformal calibration	Raw spread is not calibrated; conformal intervals have a coverage guarantee
Whole-volume accuracy vs GPU memory	Reconstruct only a 64 mm prism through the chest at the click	Keeps each ray's full path (a small box gave 73% error) at 64×128×64 voxels
Automatic detection vs false positives	The detector only suggests; the clinician clicks	The clinician stays in control; missed nodules can still be measured

3.1.1 Functional and Nonfunctional Requirements

Functional requirements state what the system must do, from loading the data to producing the report; nonfunctional requirements state the qualities it must have: speed, cost, reproducibility, validity, safety, usability, portability and privacy. Each requirement has a measurable acceptance criterion and the pipeline step that checks it (Tables 3.3 and 3.4). The criteria were chosen so that a step script can check them automatically, which makes the verification in Chapter 4 independent of anyone's judgement; Table 4.4 reports the result for every row.

Table 3.3: Functional requirements.

ID	Requirement	Acceptance criterion	Checked by
FR-01	Load LIDC-IDRI CT and radiologist annotations; build a consensus mask (≥50% of readers); split by patient into train/cal/test	DICOM and pylidc slice positions agree; splits disjoint	Steps 02–04
FR-02	Simulate low-dose DTS: 15 views, ±15°, SID 1800 mm, 43 cm panel, Poisson noise	Detector integral equals object mass within 0.1% in every view	Steps 01, 05
FR-03	Reconstruct the whole chest with SIRT, confined in depth to the patient	Residual below 0.9 of its start; missing wedge visible (depth/in-plane > 1.3)	Step 06
FR-04	Suggest nodule candidates with a learned 3D detector	Test CPM higher than a training-free baseline	Step 14

ID	Requirement	Acceptance criterion	Checked by
FR-05	Reconstruct a 64 mm prism around a click	Prism keeps the column's attenuation (ratio ≈ 1)	Step 07
FR-06	Generate eight X-ray-consistent CT hypotheses with the diffusion prior	Nodule-box PSNR of samples above the physics-only input on held-out pairs	Steps 08, 15
FR-07	Fit 3D Gaussians; measure mass, volume, 2D diameter and EUPS category	Finite values; failures reported, not hidden	Step 09
FR-08	Calibrate 90% intervals and category probabilities	Test coverage $90 \pm 3\%$; Brier score below the raw ensemble vote	Steps 10, 16
FR-09	Measure growth on virtual follow-ups (VDT, MDT, NELSON categories)	Doubling-time error and category accuracy reported	Step 11
FR-10	Explain each result: seen/unseen split, detector saliency, error drivers	Tables and figures produced on test nodules	Step 17
FR-11	Interactive click-to-3D application and HTML report of all results	Demo self-test passes; report lists every check	Steps 12, 18

Table 3.4: Nonfunctional requirements.

ID	Requirement	Acceptance criterion
NFR-01 Performance	One click gives a result quickly	≤ 60 s per nodule on one GPU
NFR-02 Compute	The study and the application run on modest hardware	One GPU for the study; free GPU for the application
NFR-03 Reproducibility	Same input, same output; checkable steps	Fixed seed, one configuration file, PASS/FAIL checks, unit tests
NFR-04 Validity	No optimistic bias	Test patients unused for training or tuning; simulation and reconstruction on different grids
NFR-05 Safety	Never report a bare number	Every number has an interval; low-confidence flag when fewer than half of the hypotheses find the nodule or the spread exceeds $10\times$
NFR-06 Usability	Usable by a clinician without coding	Browser interface, one click per nodule
NFR-07 Portability	Runs on common hardware	CPU smoke test; GPU with ≥ 24 GB for training
NFR-08 Privacy	No identifiable patient data	Only de-identified public TCIA data

3.1.2 Context Diagram

The context diagram (Fig. 3.1) depicts MedLift-3D as a single process with its external entities. The clinician is the only actor during run time: they send a case, a click and a depth and they receive the measurement report. The TCIA archive provides CT scans with the annotations from the radiologists and the cloud GPU provider performs the training and evaluation jobs and returns trained models and result, which are necessary only to build and evaluate the system, but not to use it.

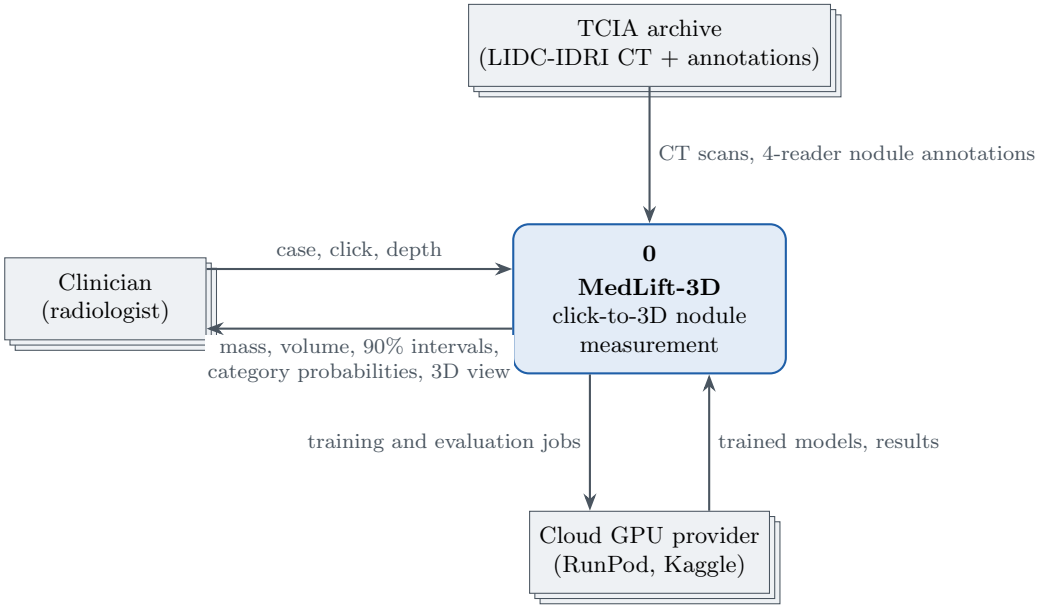


Figure 3.1: Context diagram (DFD level 0) in Gane–Sarson notation.

3.1.3 Data Flow Diagram Level 1

The level-1 DFD (Fig. 3.2) breaks down the system into eight processes and four data stores. The top row is run once for building the cases, training the models and calibrating, the second row is run every click and reads all four stores. The functional decomposition, entity-relationship model and the data dictionary are provided in Section 3.2.

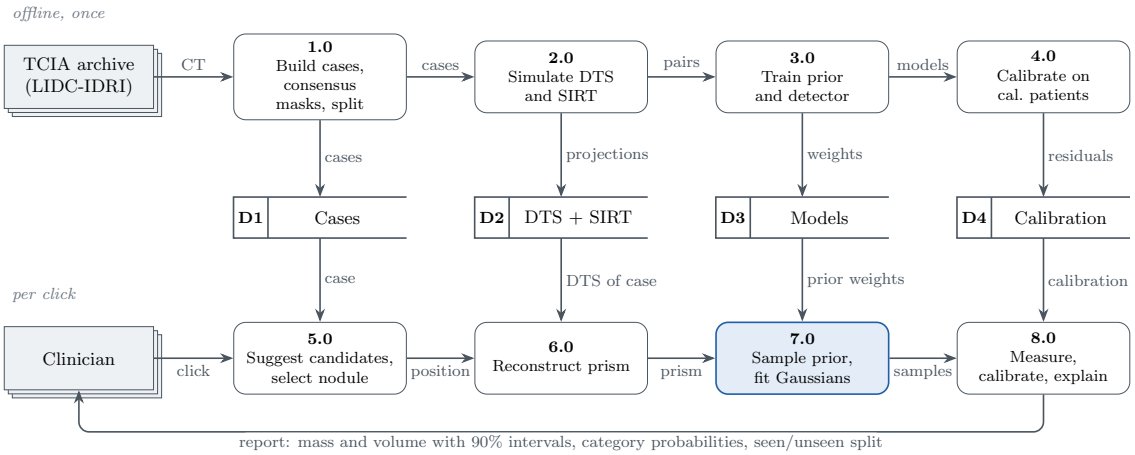


Figure 3.2: Data flow diagram level 1 (offline processes 1.0–4.0, per-click processes 5.0–8.0, data stores D1–D4).

3.1.4 UI Design

The interface is a web-based application built on a single action, which is to click on the nodule (Fig. 3.3). The numbered parts are: (1) case and method selection; (2) the coronal DTS slice, with detector suggestions, annotated nodules (for evaluation) and with the click; (3) the depth slider, which provides the third coordinate; (4) the measurement report with mass and volume intervals, the EUPS category probabilities and the low-confidence flag; (5) a 3D view of the nodule and the prism input with its eight samples; and (6) a virtual follow-up with VDT and MDT.

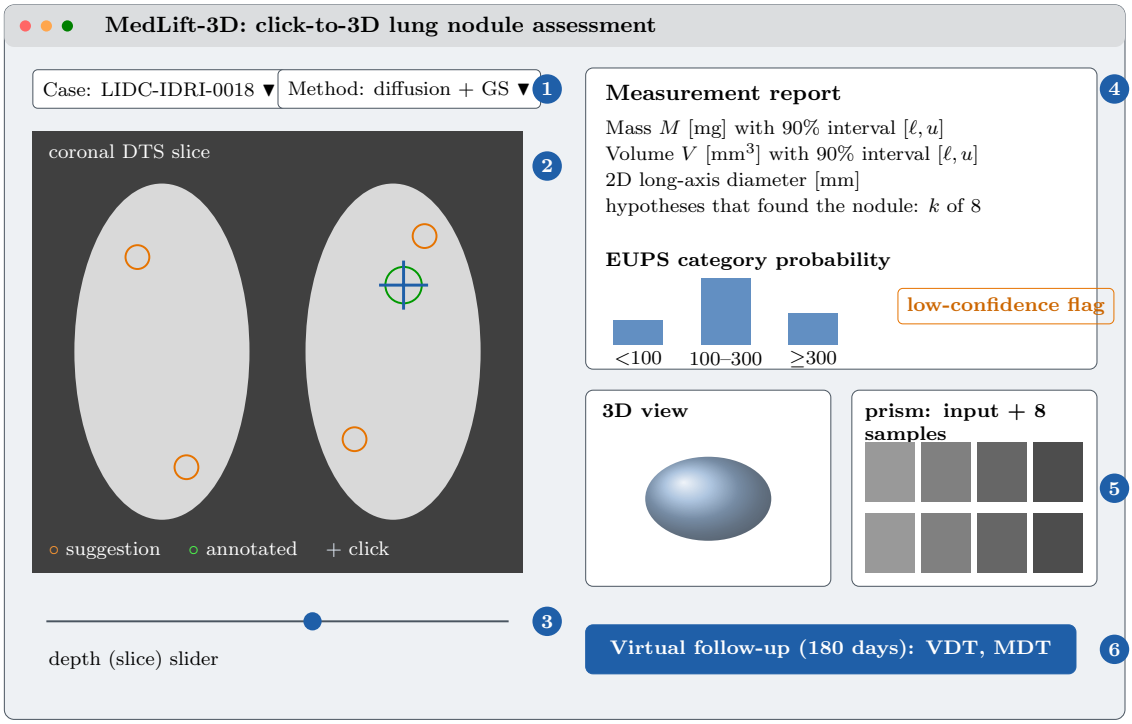


Figure 3.3: User interface design of the click-to-3D application.

3.2 Detailed Methodology and Design

This section decomposes the system into its functions and data, derives what DTS can measure, and describes the proposed architecture, the network and the alternatives considered.

3.2.1 Functional Decomposition Diagram

The system decomposes into five subsystems (Fig. 3.4). *Data and simulation* selects the patients, downloads their CT, builds the cases and simulates DTS. *Reconstruction* produces the whole-chest SIRT with depth support and the prism around a click. *Learned models* contains the conditional diffusion prior with its data-consistency sampler, the nodule detector and the fit check. *Measurement and UQ* fits Gaussians, measures mass, volume and diameter, and calibrates intervals and category probabilities. *Interface and reporting* provides the click-to-3D application, the virtual follow-up, the explanations and the HTML report. Each function maps to one or more of the 18 scripted steps, so every box can be checked by its PASS/FAIL output.

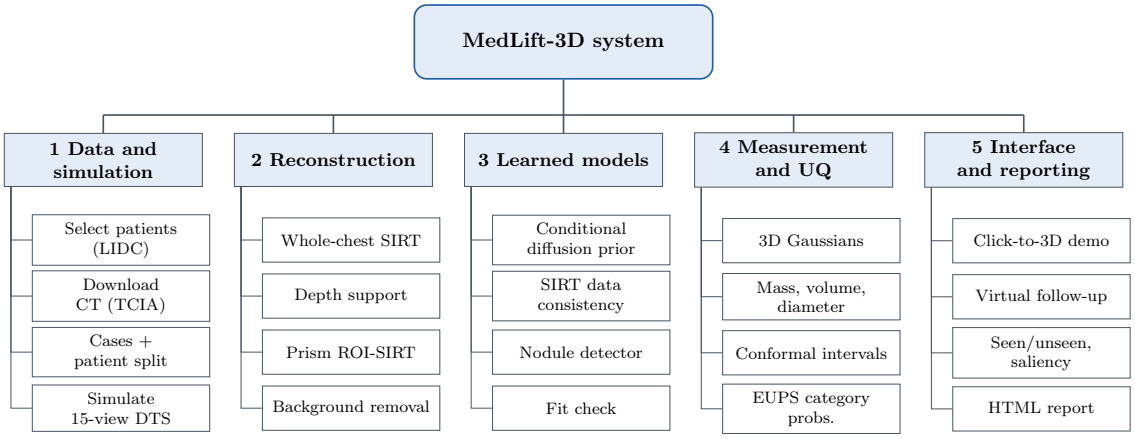


Figure 3.4: Functional decomposition diagram: five subsystems and their functions.

3.2.2 Entity-Relationship Model and Data Dictionary

Figure 3.5 shows the data entities and their relationships: a patient has CT cases, a case contains nodules marked by three or four readers, and each nodule is measured by several methods, each measurement aggregating eight hypotheses that are fitted by Gaussian sets and calibrated per method. Table 3.5 gives the data dictionary.

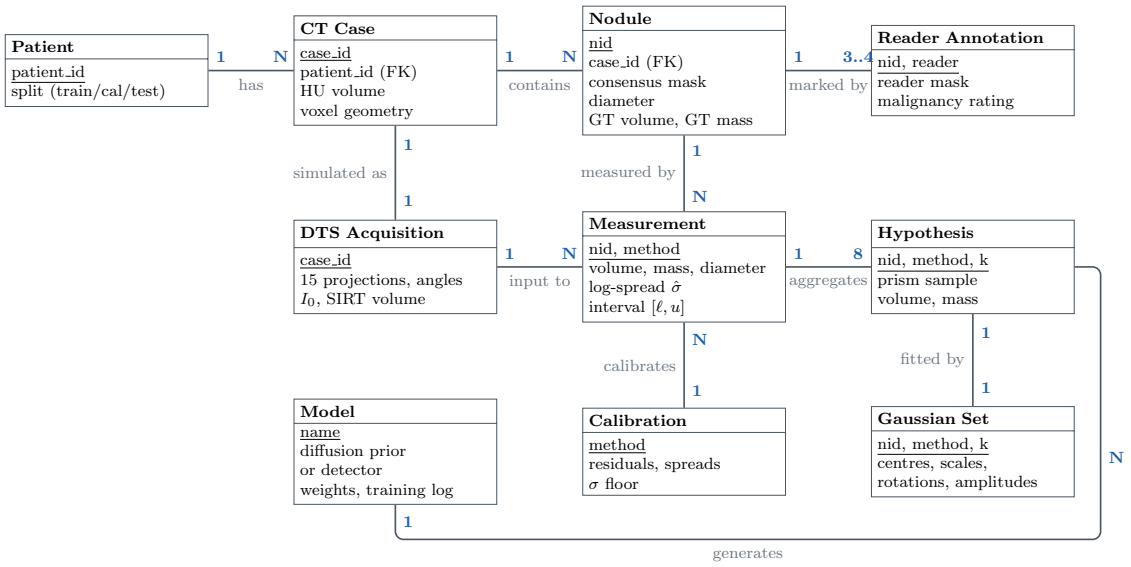


Figure 3.5: Entity-relationship model of MedLift-3D (keys underlined, cardinalities in blue).

Table 3.5: Data dictionary.

Store / item	Format	Contents	Constraint
D1 Case	int16 + packed bits + JSON header	CT in HU, consensus and four reader masks per nodule, voxel geometry, patient ID	HU in $[-1024, 3071]$
Projections	float, 15×536×536	Noisy line integrals per view	Finite, max < 15
D2 SIRT volume	float, 2.5 mm voxels	Whole-chest DTS reconstruction	Patient depth + 5 mm
Prism	float, 64×128×64, 1×2×1 mm	Column through the chest at the click	Rays $\geq$50% inside
Training pair	float16	Condition (DTS prism) and target (CT prism) in HU	Train patients only
D3 Diffusion model	PyTorch checkpoint	17.42 M parameters, EMA weights	Selected on held-out train pairs
D3 Detector	PyTorch checkpoint	0.34 M parameters, heatmap 3D U-Net	Selected on cal. patients
Gaussians	float32	Centres (N×3), scales (N×3), quaternions (N×4), amplitudes (N)	$N \leq 30{,}000$
Result row	JSON lines	Per nodule and method: volume, mass, log-spread, members, ground truth	One row per nodule
D4 Calibration	JSON	Per method: residuals, spreads, σ floor	Cal. patients only
Report	JSON / UI	Mass and volume with intervals, category probabilities, diameter, low-confidence flag, run time	Interval type stated

3.2.3 What DTS Can Measure

Let $f(x, y, z)$ be the nodule's excess attenuation, with y the depth axis, and let the view at tube angle θ project along a direction tilted by θ from y in the y–z plane, with detector coordinates (x, v) and v along $\mathbf{e}_\theta = (0, -\sin\theta, \cos\theta)$. The moments of the projection p_θ are

$$\iint p_\theta(x, v)\,\mathrm{d}x\,\mathrm{d}v = M, \tag{3.1}$$

$$\iint v\,p_\theta(x, v)\,\mathrm{d}x\,\mathrm{d}v = -\sin\theta\,M y_c + \cos\theta\,M z_c, \tag{3.2}$$

$$\iint (v - \bar{v})^2 p_\theta\,\mathrm{d}x\,\mathrm{d}v = \sin^2\theta\,S_{yy} - 2\sin\theta\cos\theta\,S_{yz} + \cos^2\theta\,S_{zz}, \tag{3.3}$$

where M is the mass, (y_c, z_c) the centroid and S the second central moments (for a cone beam, Eq. (3.1) holds up to a known magnification factor). Every view gives the mass directly (Eq. (3.1)). The depth position enters only through $\sin\theta \le \sin 15°$, making it $1/\sin 15° \approx 3.9$ times worse conditioned than in-plane position, and the depth extent S_{yy}, which volume needs, enters through $\sin^2\theta$, about $1/\sin^2 15° \approx 15$ times worse. Step 01 confirms this numerically: the detector integral matched the mass to 0.03% in all 15 views, and an object three times deeper with equal mass gave the same integrals.

3.2.4 Proposed Architecture

Figure 3.6 gives an overview of the method; its stages are described below.

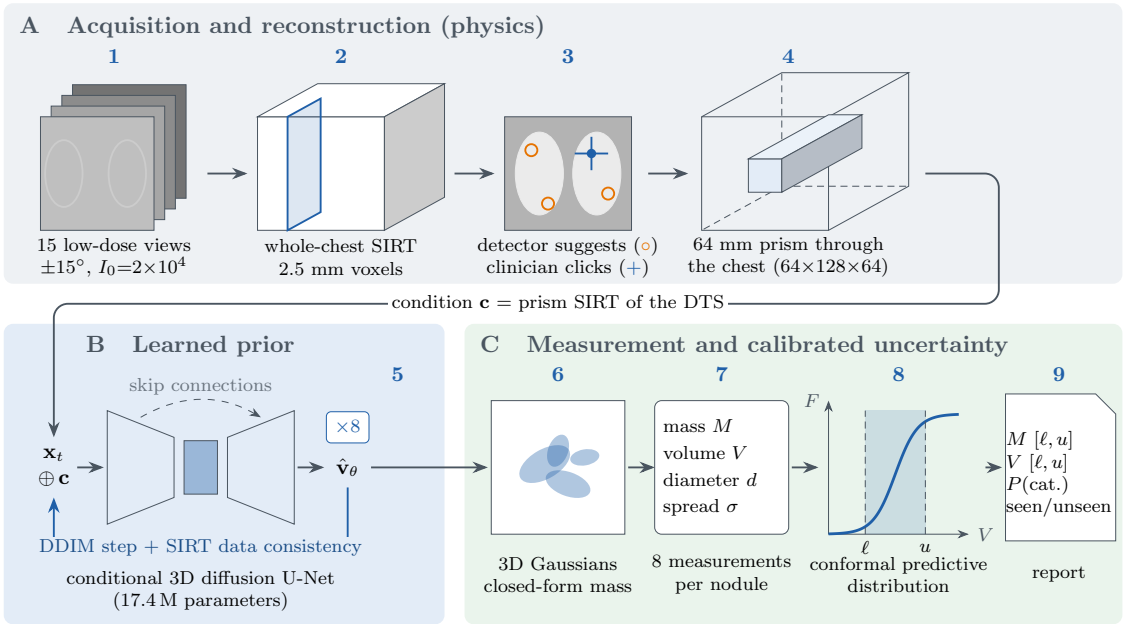


Figure 3.6: MedLift-3D method overview. (A) Physics: low-dose DTS is simulated, reconstructed and cut to a prism around the click. (B) A conditional 3D diffusion prior generates eight hypotheses of the prism, each kept consistent with the X-rays. (C) Gaussians give closed-form mass; the eight measurements are calibrated into intervals and category probabilities.

Stage 1: simulated low-dose DTS. The CT table is removed, then 15 views over $\pm 15°$ are simulated at CT resolution with 0.5 mm ray sampling, SID 1800 mm, a 43 cm flat panel with 0.8 mm pixels, monochromatic attenuation and Poisson noise at $I_0 = 2 \times 10^4$. The geometry matches the GE VolumeRAD sweep; reconstruction uses different grids than simulation to avoid the inverse crime.

Stage 2: whole-chest SIRT with depth support. SIRT (40 iterations, 2.5 mm voxels) is confined in depth to the patient (body outline + 5 mm). Without this support the reconstruction placed attenuation in the air in front of and behind the patient, and on real chests the prism lost about a third of its attenuation; with it the prism keeps 98.7%.

Stage 3: learned detector. A small 3D U-Net [54] (0.34 M parameters) predicts a nodule heatmap from the SIRT volume, and its peaks are offered as suggestions. A Laplacian-of-Gaussian and Hessian blob detector is the training-free baseline.

Stage 4: prism ROI. Around the click, a 64×64 mm column through the whole chest ($64 \times 128 \times 64$ voxels at $1 \times 2 \times 1$ mm) is reconstructed by SIRT using only rays that carry at least 50% of their attenuation inside the prism; the rest of the chest is subtracted using the coarse reconstruction. Depth is kept whole because each ray's attenuation is spread along its full path: a small 3D box around the nodule gave a 73% error against 5–7% for the prism.

Stage 5: conditional 3D diffusion prior. A 3D U-Net (Fig. 3.7; base 32 channels, multipliers 1/2/4/4, two residual blocks per level, attention at the lowest level, 17.42 M parameters) learns $p(\text{CT prism} \mid \text{DTS prism})$ with v-prediction, a cosine schedule over 1000 steps, EMA weights and bf16 mixed precision. Sampling uses 50 DDIM steps; from 60% of the way through, every denoised estimate $\hat{\mathbf{x}}_0$ receives two SIRT data-consistency iterations,

$$\hat{\mathbf{x}}_0 \leftarrow \hat{\mathbf{x}}_0 + \lambda C A^\top R\,(\mathbf{b} - A\hat{\mathbf{x}}_0), \tag{3.4}$$

before the next DDIM step. Eight samples give eight hypotheses that agree with the X-rays but differ where the X-rays are blind.

Stage 6: 3D Gaussians and measurement. For each hypothesis the nodule box ($40 \times 50 \times 40$ mm) is fitted with anisotropic Gaussians (100 fitting and 100 refinement iterations against the measured projections, the rest of the prism fixed). A seeded half-maximum segmentation, confined to a plausible nodule extent, gives the volume; Eq. (2.5) gives the mass; and the long-axis 2D diameter is kept as the clinical baseline.

Stage 7: uncertainty. The eight hypotheses give a log-scale spread $\hat{\sigma}$. On the calibration patients this spread is calibrated with (a) split-conformal asymmetric intervals (step 10) and (b) normalised and Mondrian conformal predictive systems with crepes

(step 16), which give full predictive distributions, EUPS category probabilities and category sets.

Stage 8: growth. Each nodule is grown in the CT by a known factor, its projections are updated and fresh noise added; VDT and MDT are measured and assigned to the NELSON categories.

Stage 9: explanation. The true CT prism is split into seen and unseen parts by conjugate-gradient least squares (CGLS) [55] on the prism projection operator. We measure how much of the nodule's mass and shape lies in each part and the prior's *unseen skill*, the correlation of its unseen content with the truth's (Fig. 2.1). Input×gradient saliency explains the detector, and rank correlations relate the measurement error to nodule size, depth and contrast.

3.2.5 Diffusion Network Architecture

Figure 3.7 shows the conditional 3D U-Net. The noisy prism and the DTS condition enter as two channels; four resolution levels with 32, 64, 128 and 128 channels use two residual blocks each in the encoder and three in the decoder, with skip connections between them; multi-head self-attention runs at the coarsest level (8×16×8) and in the bottleneck; and a sinusoidal timestep embedding passed through an MLP is added inside every residual block. The output layer is initialised to zero so that training starts from a stable state.

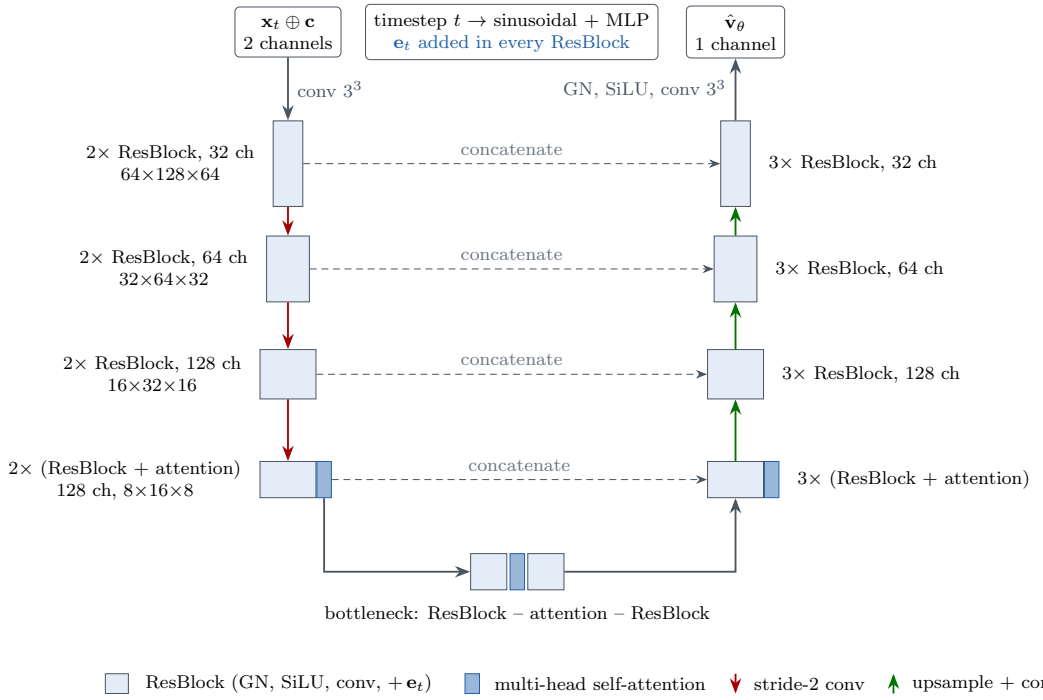


Figure 3.7: Conditional 3D U-Net of the diffusion prior. The noisy prism $\mathbf{x}_t$ is concatenated with the DTS condition $\mathbf{c}$; the timestep embedding is added in every residual block; the network predicts the velocity $\hat{\mathbf{v}}_\theta$.

3.2.6 Alternative Solutions

Table 3.6 lists the alternatives and why they were rejected. The initial design of this project generated whole 512^3 volumes with a latent diffusion model; it is included in the table because its lessons shaped the final design: a large generator judged by training losses could not show whether a single nodule measurement was right, so the goal was narrowed to clinical numbers that can be verified.

Table 3.6: Alternative solutions and the rationale for the selected design.

Alternative	Why rejected	Selected instead
FBP or SIRT alone	Cannot fill the missing wedge; physics-only volume errors of 80–90% on small nodules	SIRT as input and data-consistency step, plus a learned prior
Neural radiance fields [46]	Slow per-patient optimisation, implicit geometry, no prior for unseen depth	Explicit 3D Gaussians with closed-form mass
Whole-volume latent diffusion at 512^3 (initial design)	Too large to train and verify with our resources; evaluated by generative losses, not clinical numbers	Small conditional prior on a prism, judged by calibrated measurements
Small 3D box ROI	Each ray's attenuation lies mostly outside the box: 73% error vs 5–7%	Prism through the whole chest
GAN prior	Training instability; one output per input	Diffusion: stable training, many hypotheses
Unconditional prior with posterior sampling only	Slower and weaker for a strongly ill-posed 30° problem	Conditional model + SIRT data consistency
Raw ensemble variance or MC dropout	Not calibrated; no coverage guarantee	Ensemble spread calibrated by conformal prediction
Volume-only reporting	Volume depends on the poorly measured depth	Mass first, volume second, both with intervals

3.3 Project Plan

MedLift-3D was developed over twelve months in an iterative research and development cycle, organised in four phases that match the task groups of the Gantt chart in Fig. 3.8. The iterative approach suits the experimental nature of generative AI: the prior, the measurement and the uncertainty layer were revised whenever an intermediate verification step showed a problem.

Phase 1: Inception, data and feasibility (months 1–3). *Objective:* establish the theoretical basis and a reliable data pipeline. *Tasks:* 1.1 literature review and ethics, 1.2 dataset acquisition and preprocessing, 1.3 DTS simulation, 1.4 baseline reconstruction. *Deliverables:* literature review, curated LIDC-IDRI cases with consensus masks, the DTS simulator and baseline SIRT results.

Phase 2: Core generative architecture (months 3–6). *Objective:* build and train the learned prior. *Tasks:* 2.1 projection physics and its verification (including the projection-moment analysis), 2.2 prism ROI reconstruction, 2.3 conditional diffusion prior with data consistency, 2.4 diffusion training. *Deliverables:* verified differentiable projector, prism reconstruction, trained diffusion prior.

Phase 3: Gaussians, detection and uncertainty (months 6–9). *Objective:* turn reconstructions into calibrated, explainable measurements. *Tasks:* 3.1 Gaussian parameterisation and closed-form mass, 3.2 nodule detector, 3.3 conformal uncertainty, 3.4 explainability. *Deliverables:* measurement module, detector, calibrated intervals and probabilities, seen/unseen analysis.

Phase 4: Validation, interface and deployment (months 9–12). *Objective:* evaluate the complete system and deliver it. *Tasks:* 4.1 pilot and full-scale evaluation, 4.2 growth and sensitivity studies, 4.3 user interface development, 4.4 documentation and defense. *Deliverables:* results of the 583-patient study, the click-to-3D application, the final report.

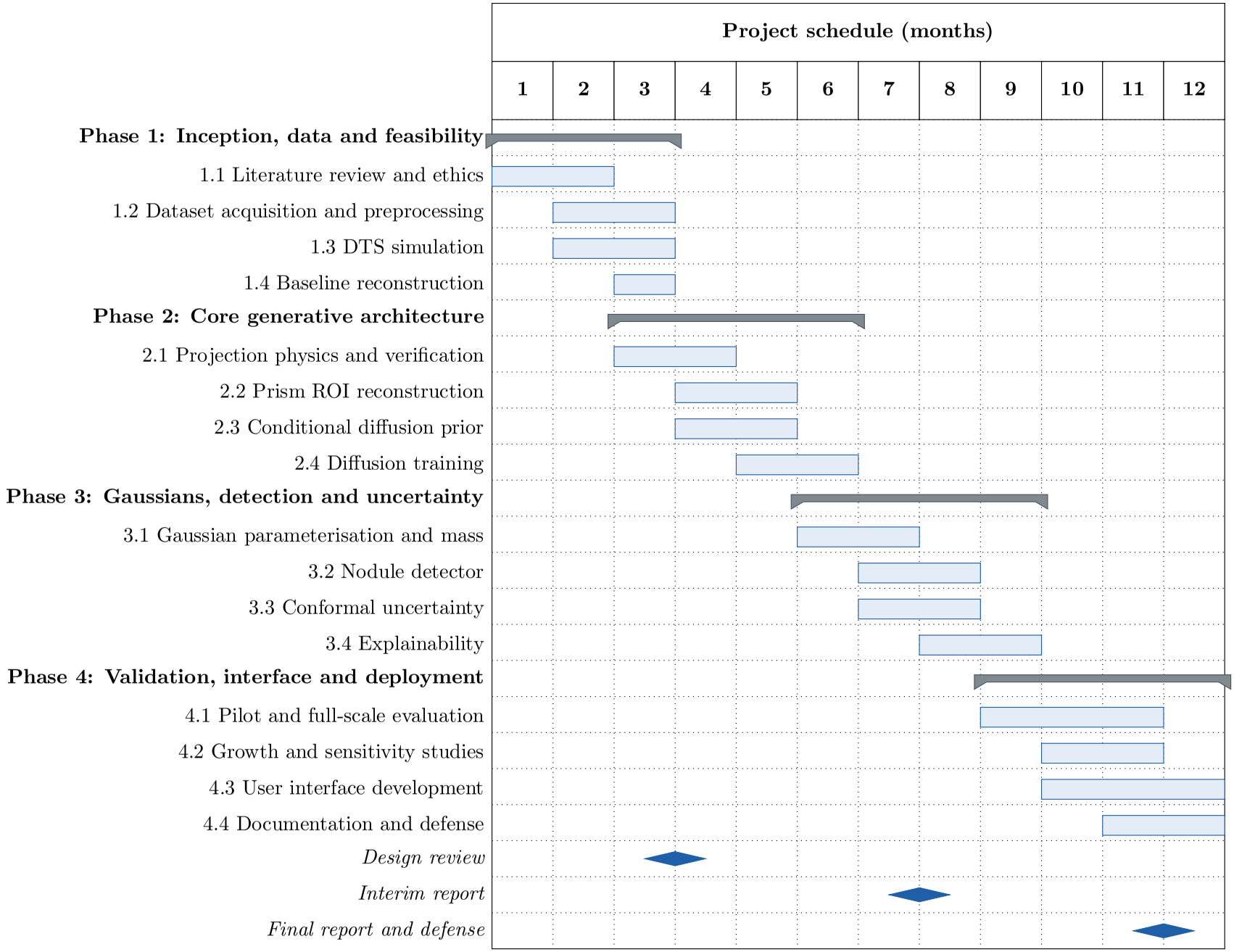


Figure 3.8: Gantt chart of the twelve-month project schedule, by phase and task.

3.4 Task Allocation

Responsibilities were allocated according to the technical area in which each member is strongest, so that the critical paths of theoretical modelling, system optimisation and application engineering could progress in parallel. The mathematically intensive parts (the diffusion prior, its conditioning and data consistency, and the statistical calibration) need a solid theoretical grounding and were led by the lead AI architect. The systems engineer focused on performance and infrastructure, making the volumetric reconstruction feasible in GPU memory and running the computation on cloud GPUs. The full-stack developer connected the algorithms to clinical use, ensuring data quality, the user interface and the evaluation outputs. Every pipeline step has one owner who is accountable for its PASS/FAIL checks, and all members reviewed every step before the full run. Table 3.7 gives the detailed allocation.

Table 3.7: Detailed task allocation matrix.

Team member	Role and focus	Detailed responsibilities
Jubair Ahmed (011222136)	Lead AI architect (theoretical modelling)	• Projection-moment analysis of what DTS can measure • Design and mathematical formulation of the conditional diffusion prior (v-prediction, cosine schedule, DDIM sampling) • SIRT data-consistency sampling to suppress anatomical hallucination • Conformal intervals and calibrated category probabilities
Foyez Ahammed Naeem (011222011)	Systems engineer (optimisation and backend)	• Differentiable DTS projector and memory-efficient prism reconstruction • GPU optimisation of training and sampling (mixed precision, batch sizing) • 3D Gaussian fitting and closed-form mass; nodule detector training • Cloud infrastructure (RunPod CPU and GPU pods, Kaggle), the full run and backups
Md Motasim Billah (011222016)	Full-stack developer (data and visualisation)	• Data curation and preprocessing of LIDC-IDRI (selection, consensus masks, DTS simulation inputs) • Clinician-facing click-to-3D interface with interactive 3D visualisation • Automated evaluation pipelines: accuracy, growth and explainability outputs, HTML report
All members	–	Literature review, verification of every PASS/FAIL step, report writing and presentation

3.5 Summary

The requirements were elicited from guidelines, the supervisor and published limits, negotiated where they conflicted, and specified with measurable acceptance criteria. The design measures mass first, reconstructs a prism around the click, fills depth with a data-consistent diffusion prior, measures with Gaussians, and calibrates and explains every number. Chapter 4 verifies each requirement.

Chapter 4

Implementation and Results

This chapter describes the hardware and tools, how every requirement was verified, and the results for hypotheses H1–H6. All numbers come from the full run of October 27, 2026 on held-out test patients unless stated otherwise.

4.1 Environment Setup

This section describes the hardware, the software tools with the reasons for choosing them, the configuration of the pipeline, and the dataset with its preprocessing.

4.1.1 Hardware

Table 4.1 lists the computing resources. CPU-only work (downloading about 64 GB of CT and building the cases) ran on a cheap CPU machine, and both machines shared one network volume so that the GPU machine started from the finished cases.

Table 4.1: Computing resources.

Resource	Used for	Specification	Access
Laptop / Kaggle CPU	Unit tests; synthetic-phantom smoke test	4 cores	Free
Kaggle notebook	40-patient pilot run; final demo and report viewer	2× NVIDIA T4, 16 GB	Free
RunPod CPU pod	Steps 02–04: patient selection, TCIA download, case building	Rented CPU	Rented
RunPod GPU pod	Steps 05–18: simulation, training, evaluation	NVIDIA L40S, 48 GB	Rented
RunPod network volume	Shared storage between pods; backup after every step	Persistent volume	Rented

4.1.2 Software and Tools

All code is written in Python, the standard language of medical AI research. PyTorch was chosen because the whole chain (projector, SIRT, diffusion network, detector and Gaus-

sian fitting) needs automatic differentiation on the GPU in one framework. Instead of an external tomography toolbox we wrote our own ray-driven DTS projector in PyTorch: it reproduces the exact linear-sweep geometry, provides gradients for the Gaussian refinement, and needs no separate CUDA build, which had caused installation problems with toolbox-based setups. LIDC-IDRI is read with pydicom and pylidc, the official tools for its annotations; conformal predictive systems use the peer-reviewed crepes library; and the application uses Gradio, which builds a browser interface directly from Python. Table 4.2 lists the tools with the alternatives considered.

Table 4.2: Software tools and the reasons for choosing them.

Tool	Role	Why chosen (alternative)
Python 3.11+	All code	Standard for medical AI (MATLAB: licence cost)
PyTorch 2.x	Projector, diffusion U-Net, detector, Gaussians; bf16 mixed precision	Differentiable physics and models in one framework
Own ray-driven projector (PyTorch)	DTS forward and back projection with gradients	Exact DTS geometry, differentiable, no CUDA build (ASTRA: separate build, no autograd)
pydicom, pylidc	LIDC DICOM and the four radiologists' annotations	Official annotation access
NumPy, SciPy, scikit-image	Morphology, segmentation, metrics	Mature and tested
crepes, scikit-learn	Conformal predictive systems; calibration metrics	Peer-reviewed implementation [29]
Gradio	Browser application	Quick UI in Python (Flask + JavaScript: more work)
Matplotlib, pytest	Figures; 69 unit tests	Standard
bash, tmux	Unattended long run with resume and backups	Survives disconnections
Git, GitHub	Version control	Team collaboration

4.1.3 Configuration and Pipeline

All of these settings (geometry, dose, prism size, network size, sampling steps, number of hypotheses, $\alpha = 0.1$) are contained in one YAML file and the random seed is fixed. There are 18 steps in the pipeline: 00 environment check, 01 physics verification, 02 patient selection, 03 downloading TCIA, 04 splitting cases, 05 DTS simulation, 06 SIRT simulation, 07 training pairs, 08 diffusion training, 09 per-nodule pipeline, 10 evaluation & conformal intervals, 11 growth, 12 demo, 13 geometry sensitivity, 14 detector, 15 fit check, 16 conformal predictive systems, 17 explainability and 18 HTML report. With each step, PASS/FAIL lines are printed, along with the expected value.

4.1.4 Dataset and Preprocessing

Source. The Lung Image Database Consortium and Image Database Resource Initiative (LIDC-IDRI) collection [19] was used, which was distributed by The Cancer Imaging Archive (TCIA) under the Creative Commons Attribution 3.0 licence. 1,018 thoracic CT scans from 1,010 patients was collected at seven academic centres and eight medical imaging companies from various manufacturers. Four expert thoracic radiologists independently reviewed each scan twice – first blinded, then with the other radiologists' markings. Each nodule 3 mm or larger was marked and each reader assessed nodule characteristics including malignancy. The only large public CT collection with nodule outlines by multiple radiologists is LIDC-IDRI, and this is precisely what is required as both a 3D ground truth and to assess inter-reader variability. A public dataset where chest DTS were paired with CT could not be found and therefore DTS were simulated from these CT scans.

Selection. Patients were selected with pylidc [20] (Fig. 4.1): a scan was kept if its slice thickness was at most 2.5 mm and it contained at least one nodule of 4–30 mm marked by at least three of the four radiologists, the convention used by LUNA16 [37]. This gave 583 patients with 1,144 nodules. Annotations from different readers were grouped into nodules by pylidc's clustering, and a nodule's *consensus mask* is the set of voxels outlined by at least 50% of its readers; the individual reader masks were kept to measure inter-reader variability.

Split. Patients, not nodules, were split 60/20/20 into training (350), calibration (117) and test (116) sets, so that no patient contributes nodules to two sets. The test set contains 225 nodules. The diffusion prior and detector were trained on the training patients, with 10% of them held out to select the diffusion checkpoint; the calibration patients were used to select the detector checkpoint and to calibrate the conformal methods; the test patients were used only for the final results.

Preprocessing. About 64 GB of DICOM was downloaded from TCIA and converted into one compact case file per scan: the CT in Hounsfield units (clipped to $[-1024, 3071]$ and stored as 16-bit integers), the consensus and reader masks of every nodule (bit-packed), and a header with the voxel geometry and patient ID, after which the DICOM was deleted. The CT table was removed by keeping the largest connected body component, because a table would not be present in a chest DTS acquisition. We verified that the DICOM slice positions agree with pylidc's for every scan (difference 0.000 mm), that the consensus masks lie on nodule tissue (median -202 HU) and that the consensus volume is on average 0.81 of the mean reader volume.

DTS simulation and training pairs. For each case, 15 projections over $\pm 15^\circ$ were simulated at CT resolution with Poisson noise ($I_0 = 2 \times 10^4$ photons per pixel and view; median single-view nodule contrast-to-noise ratio 3.3) and reconstructed by whole-chest

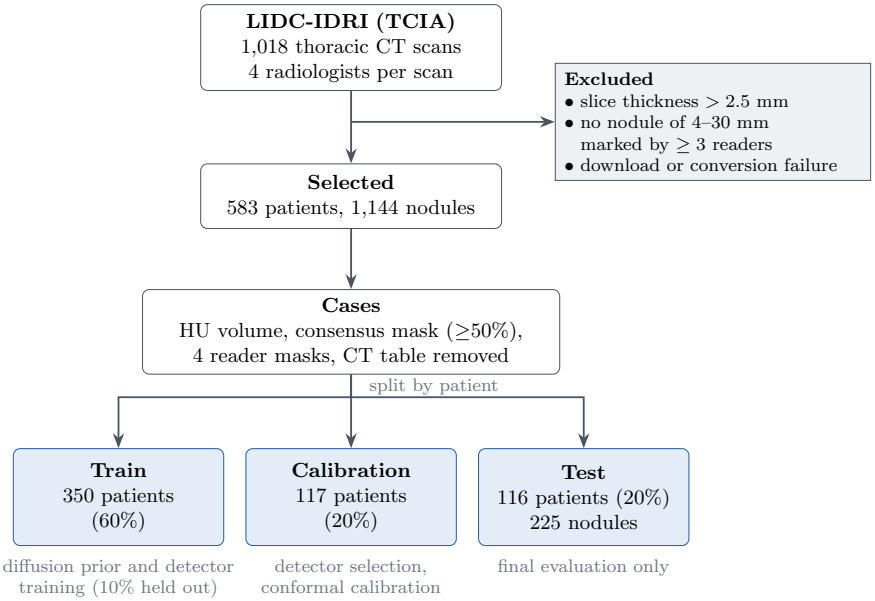


Figure 4.1: Patient selection and split of the LIDC-IDRI dataset.

SIRT. Training pairs for the diffusion prior were cut from the training patients only: for each pair, the prism SIRT of the DTS is the condition and the same prism of the CT is the target, taken around nodules and around nodule-free lung regions so that the prior also learns normal anatomy. This gave 2,522 training pairs and 265 held-out pairs from the training patients; the condition kept the prism's attenuation (median ratio to CT 0.956, 10–90%: 0.85–1.03), and its PSNR against the CT target was 10.66 dB, the baseline the prior must beat. Figure 4.2 shows the simulated views and the SIRT reconstruction of one patient and Fig. 4.4 one training pair.

Table 4.3 summarises the dataset.

Table 4.3: Dataset characteristics.

Item	Value
LIDC-IDRI patients (≥1 nodule of 4–30 mm marked by ≥3 readers, slices ≤2.5 mm)	583
Nodules	1,144
Train / calibration / test patients	350 / 117 / 116
Test nodules (detector evaluation)	225 in 116 scans
Nodules measured in step 09 (cal / test)	76 / 74
Diffusion training / held-out pairs	2,522 / 265
Radiologists' volume disagreement (median CV)	15%
Consensus volume / mean reader volume	0.81
Single-view nodule contrast-to-noise ratio (median)	3.3

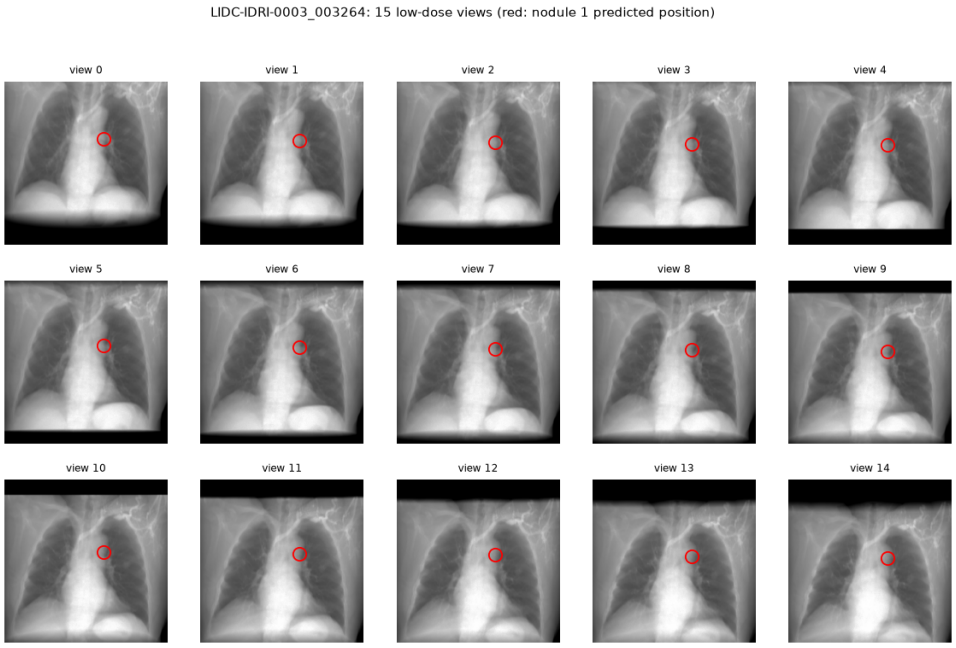


Figure 4.2: The 15 simulated low-dose views of one patient (LIDC-IDRI-0003); the red circle marks a nodule, whose shadow shifts with the tube angle.

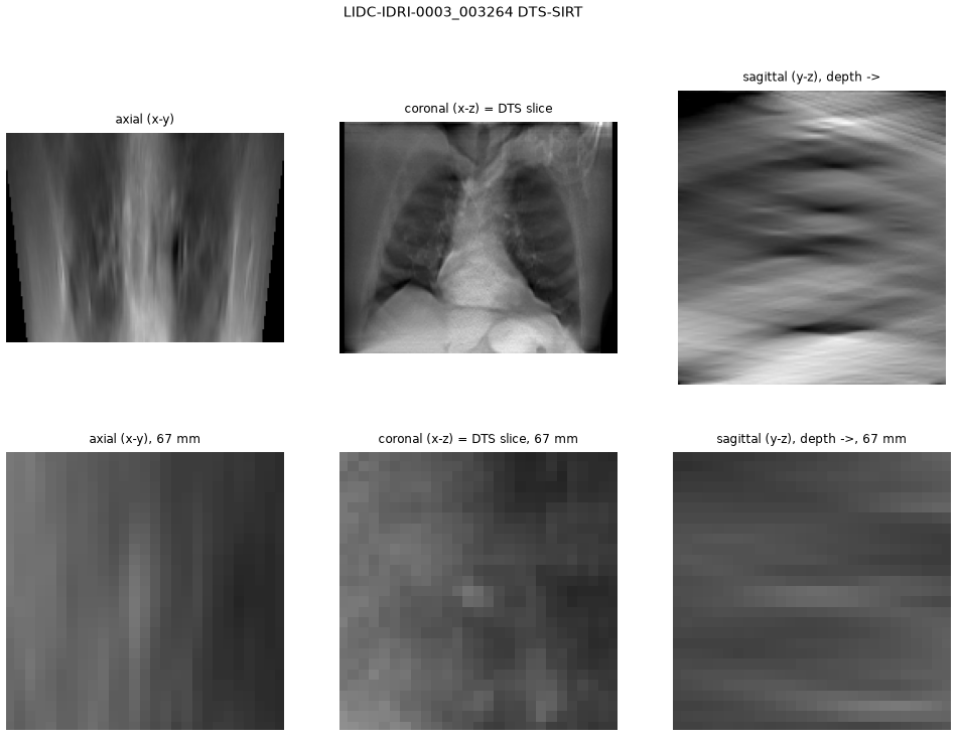


Figure 4.3: Whole-chest SIRT of the same patient: sharp in the coronal (DTS) plane, smeared in depth (sagittal).

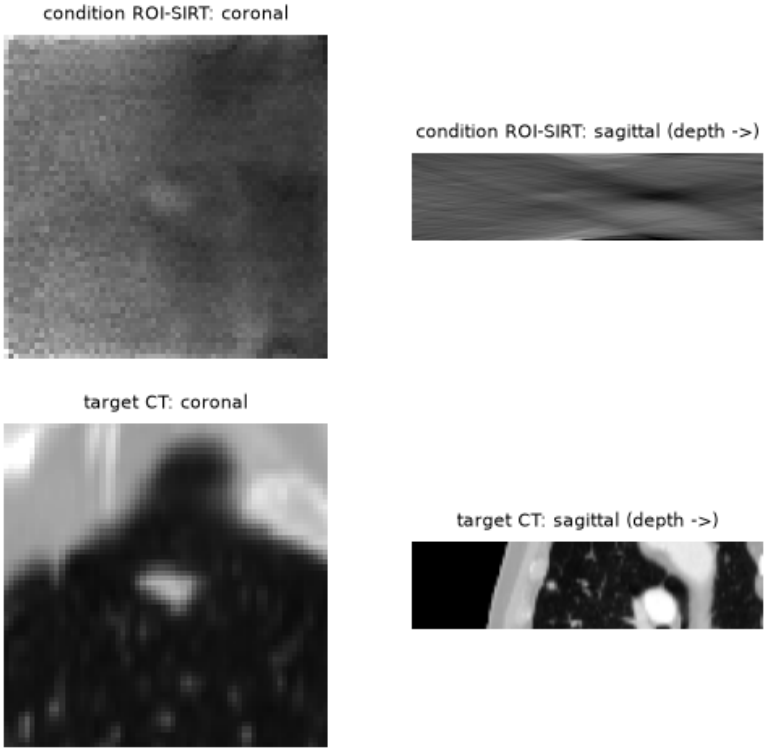


Figure 4.4: A training pair: DTS prism (condition, top) and CT prism (target, bottom), coronal and sagittal views.

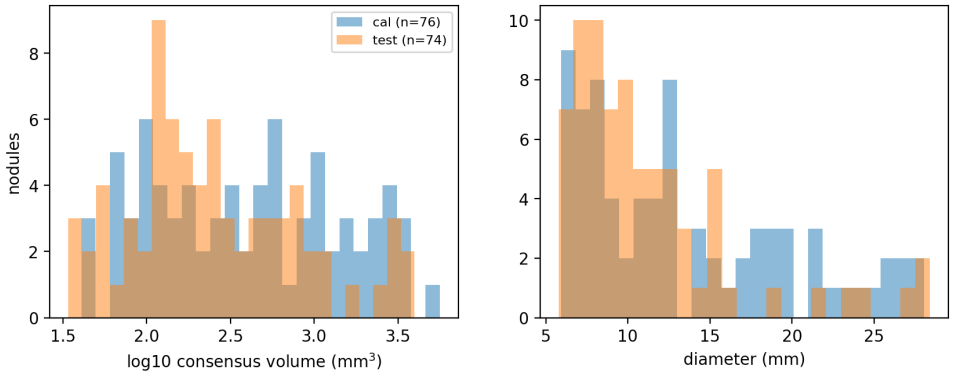


Figure 4.5: Size of the 150 measured nodules (calibration and test): consensus volume (left) and diameter (right).

4.2 Testing and Evaluation

This section outlines the testing of the system, data division, metrics/baselines, and verification of all requirements and discovery of and resolution to problems.

4.2.1 Verification Strategy

There were four levels of testing:

1. **Unit tests (69):** projector adjointness, mass conservation, conformal coverage (synthetic data), Mondrian group coverage, CPS probabilities (against crepes), CGLS seen/unseen split, and fit-check parsing.
2. **Smoke test:** entire pipeline on synthetic chest phantoms on CPU in 10-20 minutes.
3. **Real-data pilots:** CPU (4 patients) and Kaggle (40 patients) which uncovered and patched the depth-support bug (prism attenuation 67–81% $\rightarrow$ 98.7%).
4. **Full run:** 583 patients with automatic checks at each step and HTML report which contains all checks.

4.2.2 Data Split and Evaluation Protocol

Patients, not nodules, were split 60/20/20: 350 training, 117 calibration and 116 test patients. The diffusion model was selected on 10% of the training patients held out, the detector on the calibration patients, and the conformal methods were calibrated on the calibration patients. Test patients were used only for the final tables.

4.2.3 Metrics and Baselines

Volume and mass error (median signed error, median absolute percentage error MdAPE, Bland–Altman limits), failure rate, nodule-box PSNR, interval coverage and median width, Brier score and expected calibration error (ECE) of category probabilities, doubling-time error and NELSON category accuracy, detection CPM and sensitivity, seen fraction and unseen skill, and run time. Baselines: 2D diameter converted to a sphere volume (clinical practice), coarse SIRT, prism ROI-SIRT, physics-only Gaussians, and diffusion without Gaussians.

4.2.4 Requirement Verification

Each requirement in Tables 3.3 and 3.4 was checked by the step named in its last column; the check prints the measured value next to the expected one. Table 4.4 lists every requirement with its measured result. Fifteen of the sixteen rows are met. FR-08 is met for coverage but only partly for the Brier score, and the measurement accuracy itself, which no single requirement fixes, is discussed with the hypotheses in Section 4.3.

Table 4.4: Requirement verification (traceability).

Req.	Criterion	Result	Met
FR-01	Positions agree; splits disjoint	DICOM vs pylidc slice positions 0.000 mm; 350/117/116 patients, disjoint	Yes
FR-02	Integral = mass within 0.1%	0.03% in all 15 views; deeper object of equal mass indistinguishable	Yes
FR-03	Residual < 0.9; depth/in-plane > 1.3	Residual 0.02–0.03 of start; depth smear 2–3.5×	Yes
FR-04	Learned CPM > baseline	0.298 vs 0.017	Yes
FR-05	Prism keeps column attenuation	Median 0.956 of the CT (10–90%: 0.85–1.03) over all training pairs	Yes
FR-06	Samples beat physics input	Nodule-box PSNR 15.23 vs 11.88 dB (+3.35 dB)	Yes
FR-07	Finite values; failures reported	150 nodules processed without errors; failure rates reported per method (3–46%, Table 4.6)	Yes
FR-08	Coverage 90 ± 3%; Brier < vote	Coverage 95% (volume and mass); Brier 0.174–0.234 vs 0.543 for $P(V \geq 100)$, 0.284 vs 0.291 for $P(V \geq 300)$	Partly
FR-09	Growth metrics reported	Category accuracy, doubling-time error and false alarms for 40 pairs (Table 4.9)	Yes
FR-10	Explanation outputs	Seen/unseen, saliency and error-driver tables on 30 test nodules	Yes
FR-11	Demo self-test; report	Demo served from Kaggle; HTML report with 13 checked steps	Yes
NFR-01	≤ 60 s per nodule	Median 45.5 s per nodule for all five methods together on the L40S	Yes
NFR-02	One GPU; free GPU for the app	Full study on one L40S; application on a free Kaggle notebook	Yes
NFR-03	Reproducible, checkable	Fixed seed, one configuration, 69 tests, PASS/FAIL logs	Yes
NFR-04	No optimistic bias	Test patients unused until final tables; separate simulation and reconstruction grids	Yes
NFR-05	No bare numbers	Every value has an interval and a low-confidence flag	Yes

4.2.5 Issues Found and Fixed

(1) Without a depth support, SIRT placed attenuation in the air around the patient and the prism lost a third of its attenuation as false background; fixed by confining SIRT to the patient. (2) The training-free detector mostly found end-on vessels (recall in the top 10 of 6% in the pilot); replaced by the learned detector. (3) A first CPS implementation capped

probabilities at $n/(n+1)$; fixed with the smoothed mid-point p-value and verified against crepes. (4) In step 10 the asymmetric split-conformal upper bound for some methods became effectively unbounded because calibration nodules where a method found nothing (measured volume ≈ 0) produce unbounded log-ratios. This is not fixed in the reported results; its effect on the interval widths is analysed in Section 4.3, and the fix (calibrating only measurable cases) is listed as future work. (5) Three checks of the full run failed and are reported, not hidden: the H1 and H2 checks of step 10 (the hypotheses were not supported) and the virtual-growth construction check of step 11 (7.5% deviation against a 3% limit, caused by vessels near the nodule inside the mass window).

4.3 Results and Discussion

The results are presented in the order of the pipeline: the training and fit of the diffusion prior, nodule detection, measurement accuracy (H1, H2), calibrated uncertainty (H3, H5), growth (H4) and explainability (H6), followed by a discussion of why the measurements failed, the sensitivity to the acquisition geometry, and the application. The measurement results use 150 nodules (76 calibration, 74 test); detection uses all 225 test nodules.

4.3.1 Diffusion Prior Training and Fit

The prior was trained on 2,522 pairs (265 held out) for 27,610 steps (batch 8); the loss fell from 0.1186 to 0.0476 (last 10% vs first 10% of steps). On held-out pairs its samples reached 15.23 dB PSNR in the nodule box against 11.88 dB for the physics-only input (+3.35 dB) and 13.74 vs 10.16 dB over the whole prism (Fig. 4.6). The fit check (Fig. 4.7) gives two messages. The denoising loss at fixed noise levels was nearly the same on training and held-out pairs (0.0342 vs 0.0331, ratio 0.97), so the network did not memorise its inputs; but the sample quality gain was larger on training patients than on held-out ones (+7.00 vs +4.22 dB), so its samples are better for patients it has seen. The held-out gain was still rising at the end (+1.89 to +3.35 dB over the last checks), so the prior is also under-trained.

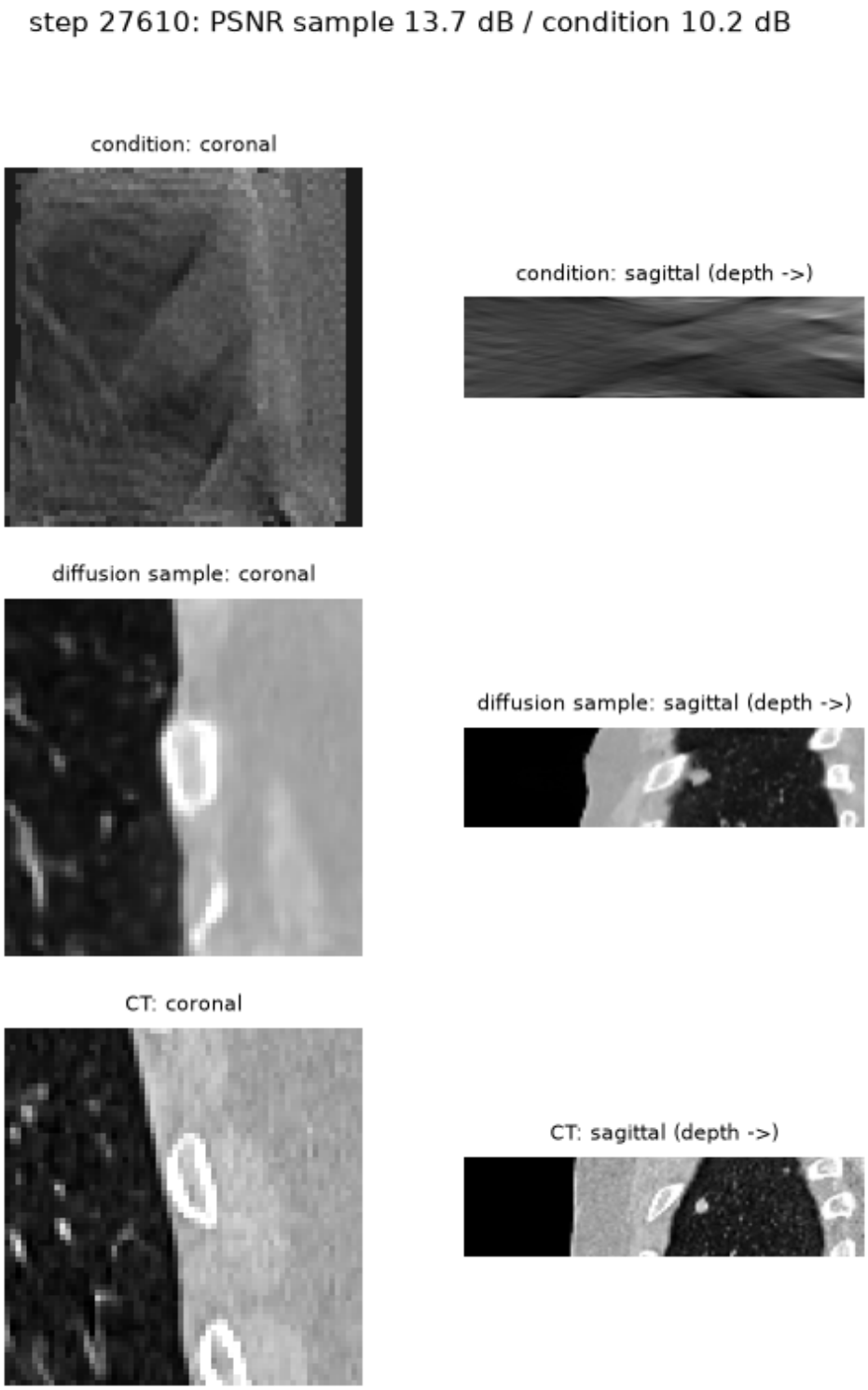


Figure 4.6: A held-out training pair at the end of training: DTS condition (top), diffusion sample (middle) and CT (bottom), coronal and sagittal (depth) views. The sample has realistic anatomy and sharp depth structure.

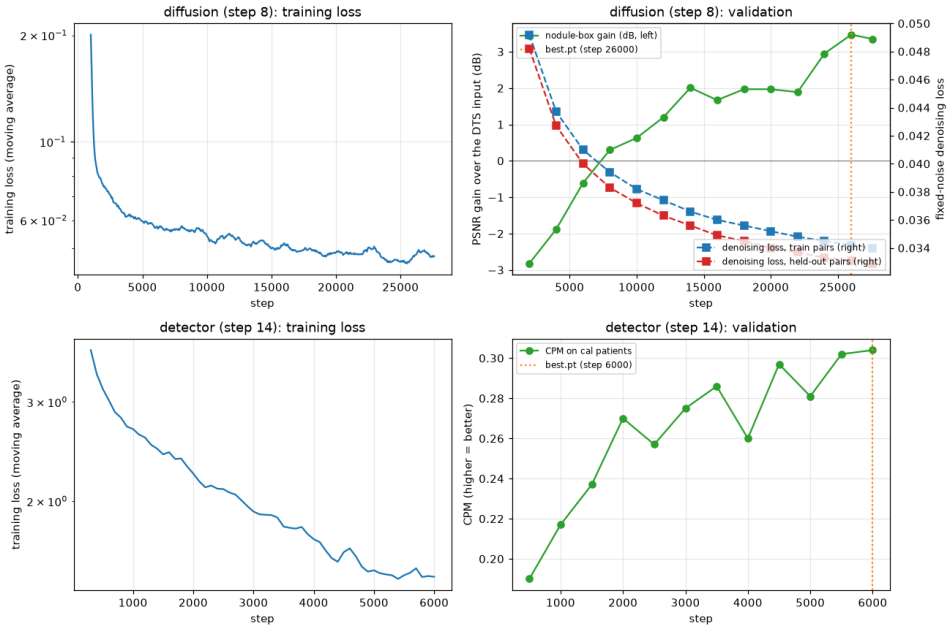


Figure 4.7: Fit check (step 15): training loss and validation of the diffusion prior (top) and of the detector (bottom).

4.3.2 Nodule Detection

Table 4.5: Nodule detection on the 116 test patients (225 nodules).

Detector	CPM	Sensitivity at 1 / 2 / 4 / 8 FP per scan	Recall in top 10
Learned 3D U-Net (ours)	**0.298**	28% / 35% / 40% / 46%	**48%**
Training-free LoG + Hessian	0.017	0% / 1% / 3% / 7%	8%
Learned, in-plane position only (diagnostic)	0.300	28% / 35% / 40% / 47%	49%

The learned detector is 17 times better than the training-free baseline by CPM (Table 4.5 and Fig. 4.8). The in-plane-only row is almost identical, so its misses are not suggestions at the wrong depth but nodules that a DTS view does not separate from vessels and chest wall. Its validation CPM had reached a plateau (0.304 at step 6,000), but it scored 0.484 on training patients against 0.329 on calibration patients, so it partly memorised its 350 training patients; more patients, not more steps, would help. A sensitivity of 35% at two false positives per scan is far below CT detectors, which is why the detector only suggests and the clinician can click anywhere.

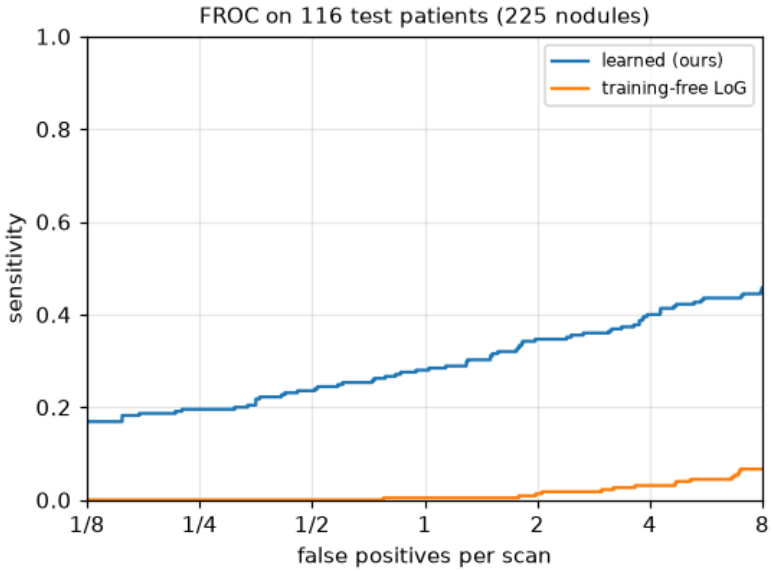


Figure 4.8: FROC curves of the learned and training-free detectors on the test patients (step 14).

4.3.3 Measurement Accuracy (H1, H2)

Each test nodule was measured by six methods: the clinical 2D diameter converted to a sphere volume, the coarse whole-chest SIRT, the prism ROI-SIRT, physics-only Gaussians, the diffusion prior without Gaussians, and the full method (diffusion prior with Gaussians). Volumes are compared with the radiologists' consensus volume and masses with the CT mass inside the consensus mask plus a 1 mm rim. Table 4.6 reports the failure rate (nothing found at the click), the median signed error (bias) and the median absolute percentage error (MdAPE), in which a failure counts as a 100% error.

Table 4.6: Measurement error on the 74 test nodules (step 10). A failure counts as 100% error in the MdAPE.

	Volume				Mass			
Method	Failed	Bias	MdAPE	Within ±20%	Failed	Bias	MdAPE	Box PSNR
2D diameter → sphere (clinical)	15%	−44%	88.0%	7%	–	–	–	–
Coarse DTS SIRT	46%	−61%	100.0%	5%	46%	−66%	100.0%	12.28
Prism ROI-SIRT (physics)	15%	+10%	91.2%	**14%**	16%	+8%	100.0%	12.14
Physics-only Gaussians	7%	−82%	87.7%	3%	7%	−43%	**78.0%**	11.99
Diffusion only	**3%**	−58%	**74.5%**	9%	36%	−82%	100.0%	**17.49**
Diffusion + Gaussians (ours)	22%	−91%	94.6%	4%	22%	−87%	97.3%	14.72

H1 (mass error < volume error) is not supported. Mass had a lower MdAPE than volume only for the physics-only Gaussians (78.0% vs 87.7%); for the other methods it was equal or worse. **H2 (the prior reduces volume error) is not supported for the full method**: diffusion with Gaussians had a volume MdAPE of 94.6% against 87.7–91.2% for the physics-only methods. The diffusion prior alone had the lowest volume MdAPE (74.5%) and the fewest failures (3%), but a strong negative bias (−58%). No method reached clinically useful accuracy: even the best had only 14% of nodules within ±20%,

and the clinical 2D baseline had an MdAPE of 88%. For comparison, the radiologists' own volumes differ by a median coefficient of variation of 15%.

Two observations explain these numbers. First, image quality and measurement accuracy disagree: the diffusion prior raised the nodule-box PSNR from about 12 dB to 17.49 dB, yet its volumes are the most biased. Second, all learned methods *under*-estimate: Figs. 4.9 and 4.10 show most diffusion and Gaussian estimates far below the line of identity, and Fig. 4.11 shows that the full method's errors cluster at -100% (nodules shrunk or erased) with a few large over-estimates. Figure 4.12 shows the error distributions of volume and mass side by side and the volume error against nodule size: errors are large at every size.

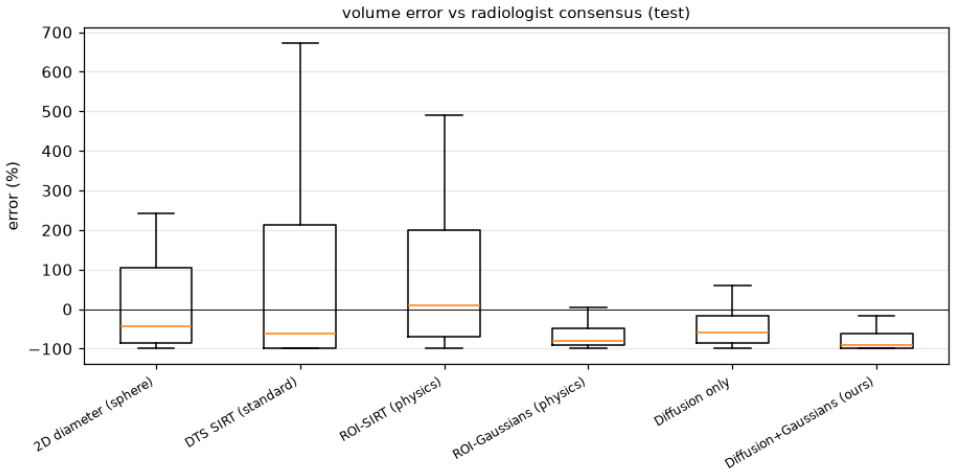


Figure 4.9: Volume error of each method against the radiologist consensus on the test nodules (step 10).

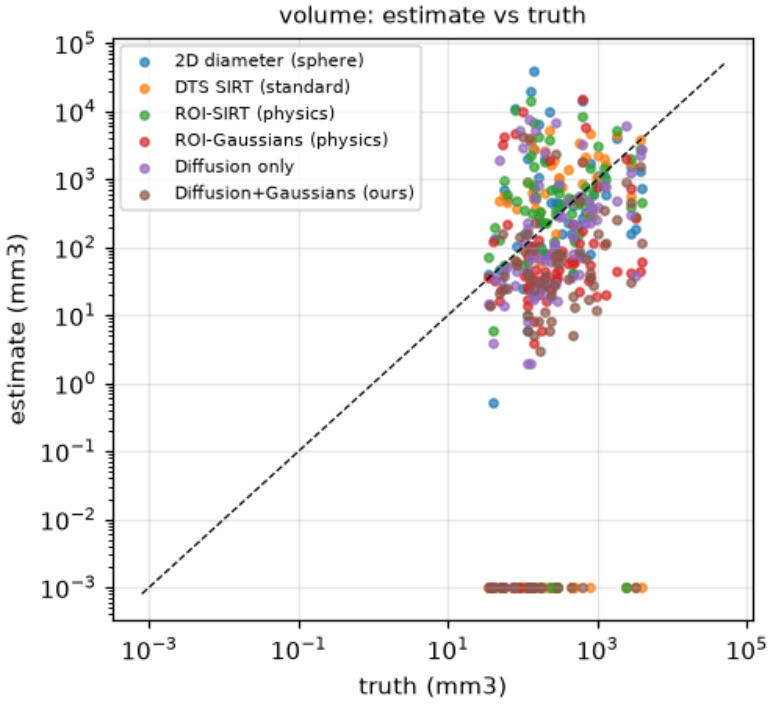


Figure 4.10: Estimated vs consensus volume for all methods (test nodules); failures are shown at 10^{-3} mm^3.

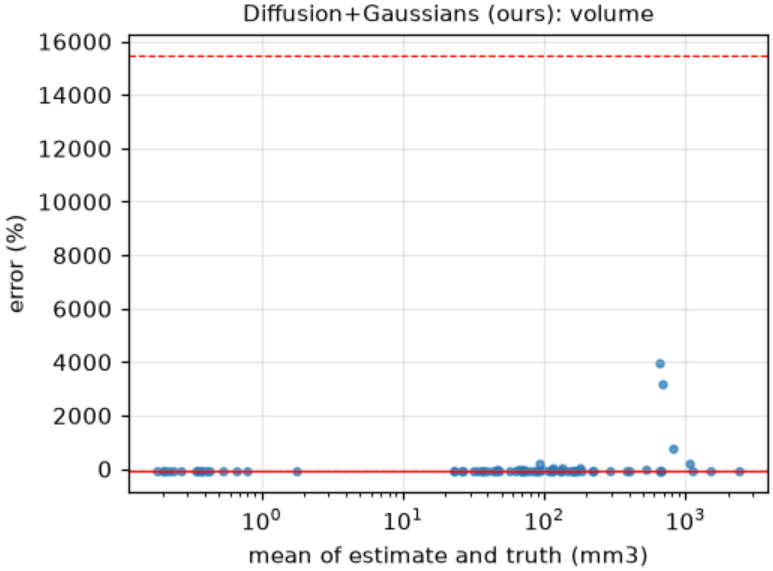


Figure 4.11: Bland–Altman plot of volume for diffusion with Gaussians (test nodules).

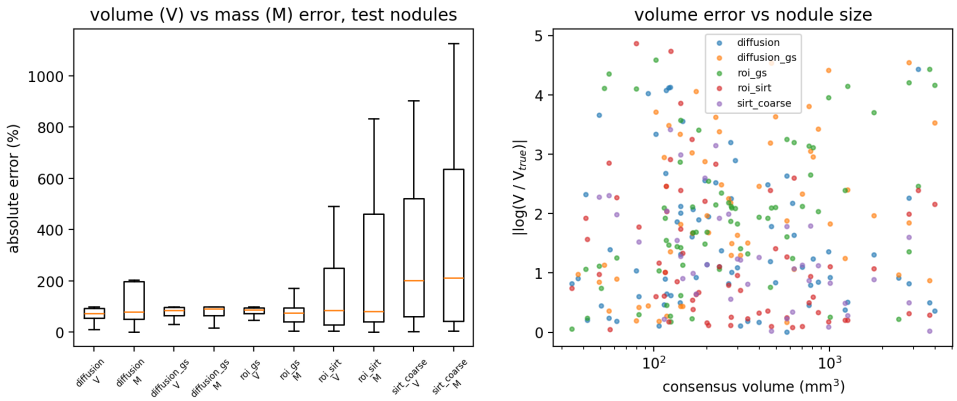


Figure 4.12: Left: absolute error of volume (V) and mass (M) for each method (test nodules, failures excluded). Right: volume error against nodule size.

4.3.4 Calibrated Uncertainty (H3, H5)

Table 4.7 gives the conformal intervals of the full method. All variants calibrated on the 76 calibration nodules reach or exceed the 90% target coverage on the test nodules except the Mondrian variant for volume, so the coverage guarantee of conformal prediction held in practice (Fig. 4.14). The raw ensemble spread, used as $\pm z$ standard deviations, covered only 28% at the 90% level. **However, the intervals are not informative.** Their median width is many orders of magnitude larger than the estimate, and the normalised variants are effectively unbounded. This is because failures: if a measurement is zero then the score is infinite, and with 22–26% failure on the calibration nodules the conformal quantile score is infinite. The only exception is the Mondrian variant which has finite widths, at about 58 times the estimate. H3 thus has validity (coverage) but not usefulness (width) and the sub-claim that mass intervals are narrower than volume intervals can be determined only with bounded widths. Both results are shown for each test nodule (Fig. 4.13): the truth is contained within 70 of the 74 intervals; however, 61 upper bounds of the intervals are above 10^5 mm^3 and 16 of the failed reads (estimate zero) are at the bottom of their intervals.

Table 4.7: Conformal 90% intervals of diffusion with Gaussians on the 74 test nodules (step 16). Width = (upper − lower) / estimate.

	Volume			Mass		
Variant	Coverage	Median width	Worst size group	Coverage	Median width	Worst size group
Standard	96%	1.2×10^8%	88%	96%	5.1×10^7%	89%
Normalised by ensemble spread	95%	unbounded	82%	95%	unbounded	89%
Mondrian size groups	78%	**5784%**	67%	92%	**6208%**	89%
Mondrian + normalised	88%	4.9×10^6%	79%	97%	3.2×10^7%	**96%**

The category probabilities are read off each nodule's calibrated predictive distribution, the probability that its volume is at most v for every v (Fig. 4.16): $P(V \geq 100)$ is one minus its value at 100 mm^3. **H5 is partly supported.** Calibration clearly helped the probability that a nodule is at least 100 mm^3: the Brier score fell from 0.543 (raw

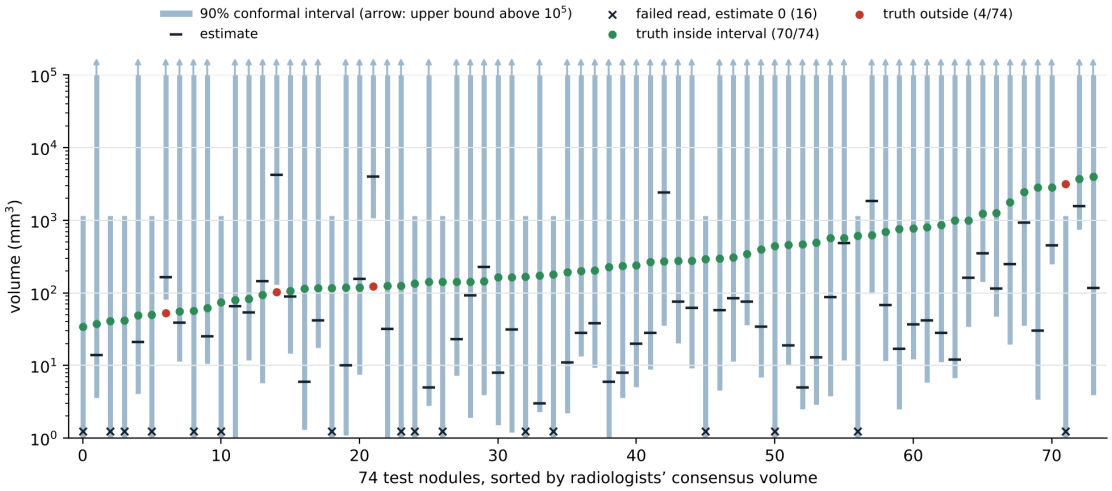


Figure 4.13: Conformal 90% volume intervals (normalised by the ensemble spread) for the 74 test nodules, sorted by the radiologists' consensus volume. Green: truth inside the interval; red: outside; cross: failed read with estimate zero; arrow: upper bound beyond the plotted range.

Table 4.8: EUPS category probabilities on the test nodules (step 16). Lower Brier score and ECE are better.

Probabilities from	Brier $P(V \geq 100)$	ECE $P(V \geq 100)$	Brier $P(V \geq 300)$	ECE $P(V \geq 300)$	Most likely category correct
Conformal (normalised)	0.234	0.206	**0.284**	**0.236**	**38%**
Conformal (Mondrian + normalised)	**0.174**	**0.115**	0.336	0.257	**38%**
Ensemble vote (uncalibrated)	0.543	0.583	0.291	0.253	26%
Ensemble spread as log-normal (uncalibrated)	0.526	0.561	0.295	0.275	27%
Point estimate only (0/1)	0.635	0.635	0.324	0.324	27%

vote) to 0.174 and the ECE from 0.583 to 0.115 (Table 4.8 and Fig. 4.15), and the most likely category was right for 38% of nodules instead of 26%. For the 300 mm^3 threshold calibration made almost no difference (0.284 vs 0.291). The category sets from the 90% intervals contained the true category for 99% of nodules but held 2.91 of the 3 categories on average, again valid but uninformative. Mondrian calibration raised the worst size-group coverage for mass from 89% to 96% but not for volume (82% to 79%; Fig. 4.17), so size-conditional coverage was achieved for mass only.

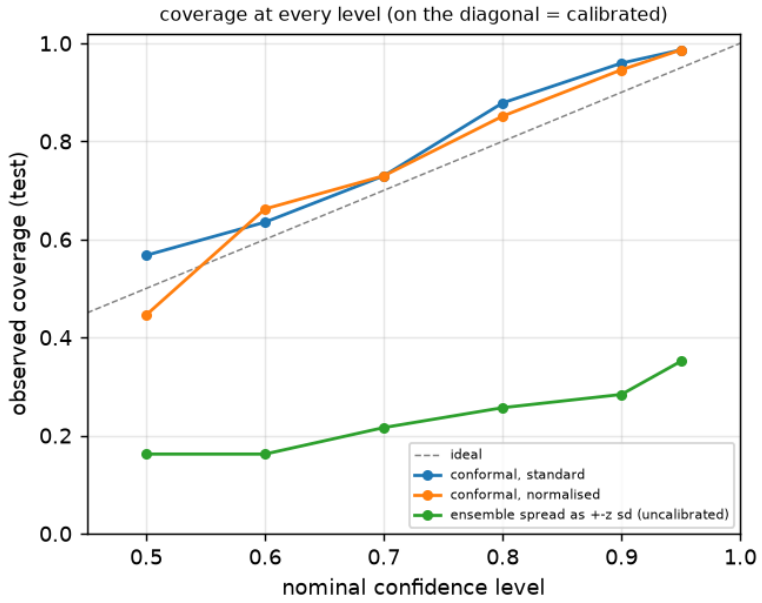


Figure 4.14: Observed coverage at every nominal level for the volume intervals (test nodules): conformal intervals lie on or above the diagonal, the raw ensemble spread far below it.

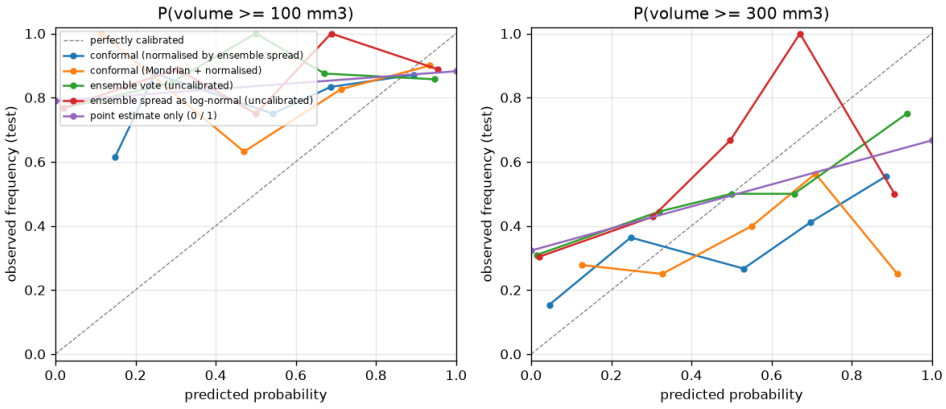


Figure 4.15: Reliability diagrams of the EUPS category probabilities for $P(V \geq 100)$ and $P(V \geq 300)$ (test nodules).

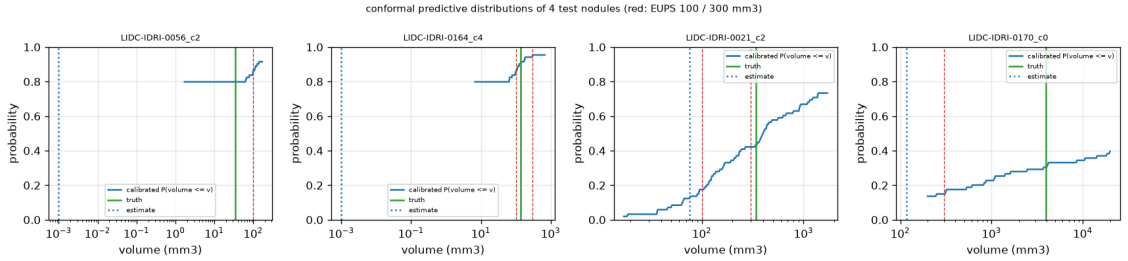


Figure 4.16: Calibrated conformal predictive distributions of four test nodules: the probability that the volume is at most v (blue), the estimate (dotted), the truth (green) and the EUPS thresholds of 100 and 300 mm^3 (red). A steep curve near the truth is a confident, correct prediction; a shallow one spreads the probability over all categories.

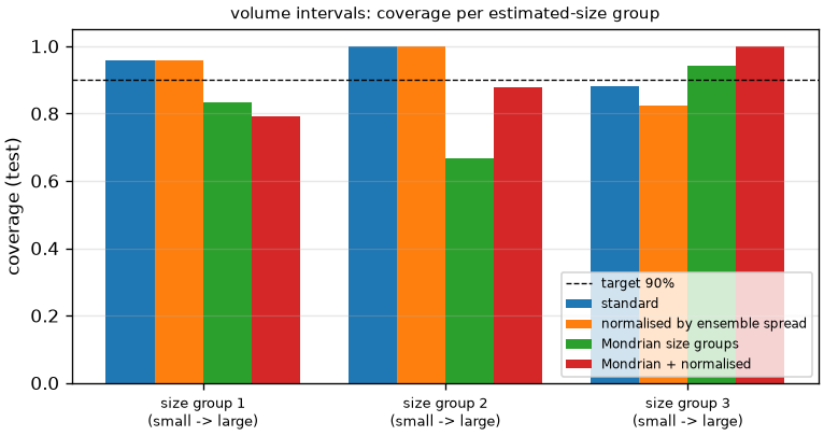


Figure 4.17: Coverage of the volume intervals in three size groups, with and without Mondrian calibration.

4.3.5 Growth (H4)

For the growth experiment each nodule was enlarged inside the CT by a known factor, corresponding to a known true doubling time, and a follow-up DTS acquisition was simulated with fresh noise; 40 growth pairs were evaluated. VDT and MDT were computed from the volume and mass ratios and assigned to the NELSON categories (< 400, 400–600 and > 600 days).

Table 4.9: Doubling-time results on 40 virtual growth pairs (step 11).

Method	Quantity	Category accuracy	Median $\lvert\log_2(\mathrm{DT_{est}}/\mathrm{DT_{true}})\rvert$ (growing)	False growth alarms (stable)	Not measurable
Coarse SIRT	VDT / MDT	35% / 32%	2.14 / 0.88	0% / 0%	35% / 35%
Prism ROI-SIRT	VDT / MDT	48% / **50%**	0.82 / **0.77**	10% / **0%**	18% / 18%
Physics-only Gaussians	VDT / MDT	42% / 40%	1.13 / 1.29	20% / 20%	10% / 10%
Diffusion only	VDT / MDT	48% / 28%	0.91 / 0.96	30% / 30%	5% / 48%
Diffusion + Gaussians (ours)	VDT / MDT	30% / 30%	1.23 / 1.11	10% / 20%	40% / 40%

H4 is not supported for our method (MDT and VDT both 30% category accuracy, with 40% of pairs not measurable). For the physics-only prism reconstruction, MDT was slightly better than VDT (50% vs 48% accuracy, doubling-time error 0.77 vs 0.82 in $\log_2$ units, no false growth alarms), and for the coarse SIRT the MDT error was much smaller than the VDT error (0.88 vs 2.14), in line with the mass-first argument; but these differences are small for 40 pairs. One verification check of this step failed: the mass added by the virtual growth deviated from its intended value by up to 7.5% (limit 3%), because the mass window also contains nearby vessels; the growth results are therefore indicative only. Figure 4.18 shows estimated against true doubling times.

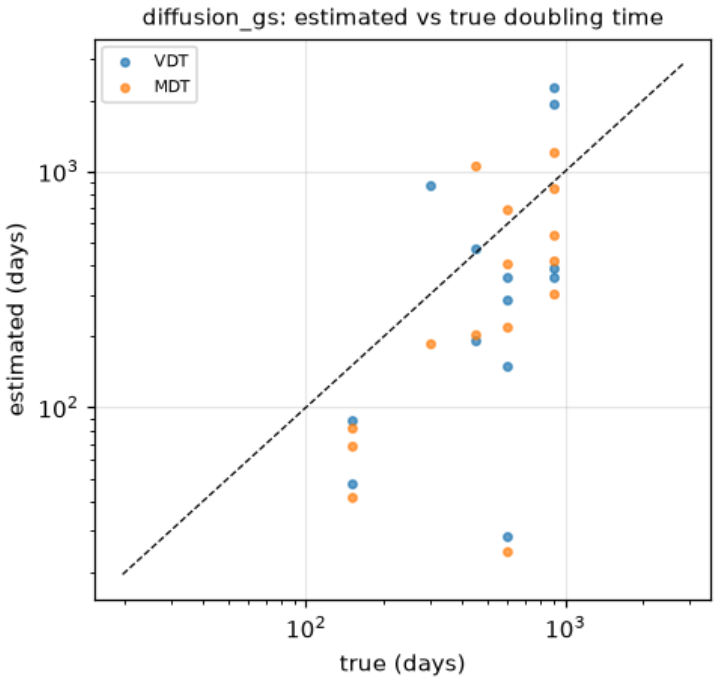


Figure 4.18: Estimated vs true doubling time for diffusion with Gaussians (VDT and MDT).

4.3.6 Explainability (H6)

Table 4.10: Seen/unseen analysis inside the nodule box, median [IQR] over 30 test nodules (step 17). Unseen skill: 1 = the invisible part matches the truth, 0 = no better than leaving it out, < 0 = invented structure.

Volume	Unseen share of box signal	Unseen share of box mass	Unseen skill vs truth	Seen skill vs truth
CT truth	0.15 [0.09, 0.27]	0.00 [0.00, 0.00]	–	–
Prism ROI-SIRT (physics)	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.01]	0.39 [0.19, 0.56]
Diffusion (mean of samples)	0.09 [0.05, 0.15]	0.00 [0.00, 0.00]	**0.22 [0.14, 0.36]**	**0.88 [0.75, 0.93]**
Diffusion + Gaussians (ours)	0.04 [0.02, 0.07]	0.00 [0.00, 0.00]	0.09 [0.03, 0.19]	0.59 [0.39, 0.74]

The split itself is accurate (median $|A\mathbf{x}_{\text{unseen}}|/|A\mathbf{x}| = 0.002$). **The analysis confirms the physics behind H1**: only 0.3% of the true nodule's mass lies in the unseen part, while 15% of the nodule-box signal (its shape in depth) is invisible to the 15 views. **H6 is supported for the diffusion prior**: its unseen content has a positive skill of 0.22 [0.14, 0.36] against the true CT, so it recovers part of what the views miss rather than inventing structure, and it reproduces the seen part far better than the physics-only reconstruction (0.88 vs 0.39). After Gaussian fitting the skill drops to 0.09, so the Gaussian step discards much of what the prior recovered. Nodules that are less visible to DTS had larger volume errors (Spearman -0.47 [-0.75, -0.12], $n = 30$). Figure 4.19 shows one nodule. Figure 4.20 shows an 11.6 mm nodule of the same patient where the prior works as intended and the Gaussian step then fails: the diffusion samples recover the hidden depth (unseen skill 0.36, seen skill 0.93), but the Gaussian fit returns no nodule at all (volume zero against 615 mm^3), the failure mode traced in the section on why the measurements failed.

Table 4.11: Error drivers for diffusion with Gaussians: Spearman correlation with the absolute error [95% CI], 150 nodules (step 17).

Nodule property	Volume error	Mass error
Diameter (radiologists)	-0.10 [-0.26, $+0.07$]	-0.20 [-0.34, -0.03]
Solidity (LIDC texture)	-0.21 [-0.36, -0.05]	-0.17 [-0.32, -0.01]
Density (mean HU)	-0.32 [-0.46, -0.17]	-0.35 [-0.49, -0.20]
Radiologists' disagreement (CV)	$+0.04$ [-0.13, $+0.20$]	$+0.09$ [-0.09, $+0.27$]
Malignancy rating	-0.01 [-0.19, $+0.16$]	-0.12 [-0.28, $+0.04$]
Ensemble spread	$\mathbf{-0.52}$ [-0.65, -0.37]	$\mathbf{-0.50}$ [-0.64, -0.35]
Physics data misfit	-0.08 [-0.24, $+0.08$]	-0.04 [-0.19, $+0.12$]

Table 4.11 shows which nodules fail. Faint (low-density) and less solid nodules have larger errors, while size and the radiologists' own disagreement matter little. The most important finding is the sign of the ensemble spread: it is *negatively* correlated with the error (-0.52). The eight samples agree most when they are most wrong, which happens when they all erase a faint nodule. The spread therefore signals the opposite of what an uncertainty should, which is also why the intervals normalised by it became unbounded.

For the detector, the share of input×gradient saliency within 10 mm of the suggestion

was 0.15 for true positives and 0.16 for false positives (Fig. 4.21): the detector looks at the local region in both cases, and false positives come from local structures that resemble nodules, such as vessels seen end-on, not from attention elsewhere.

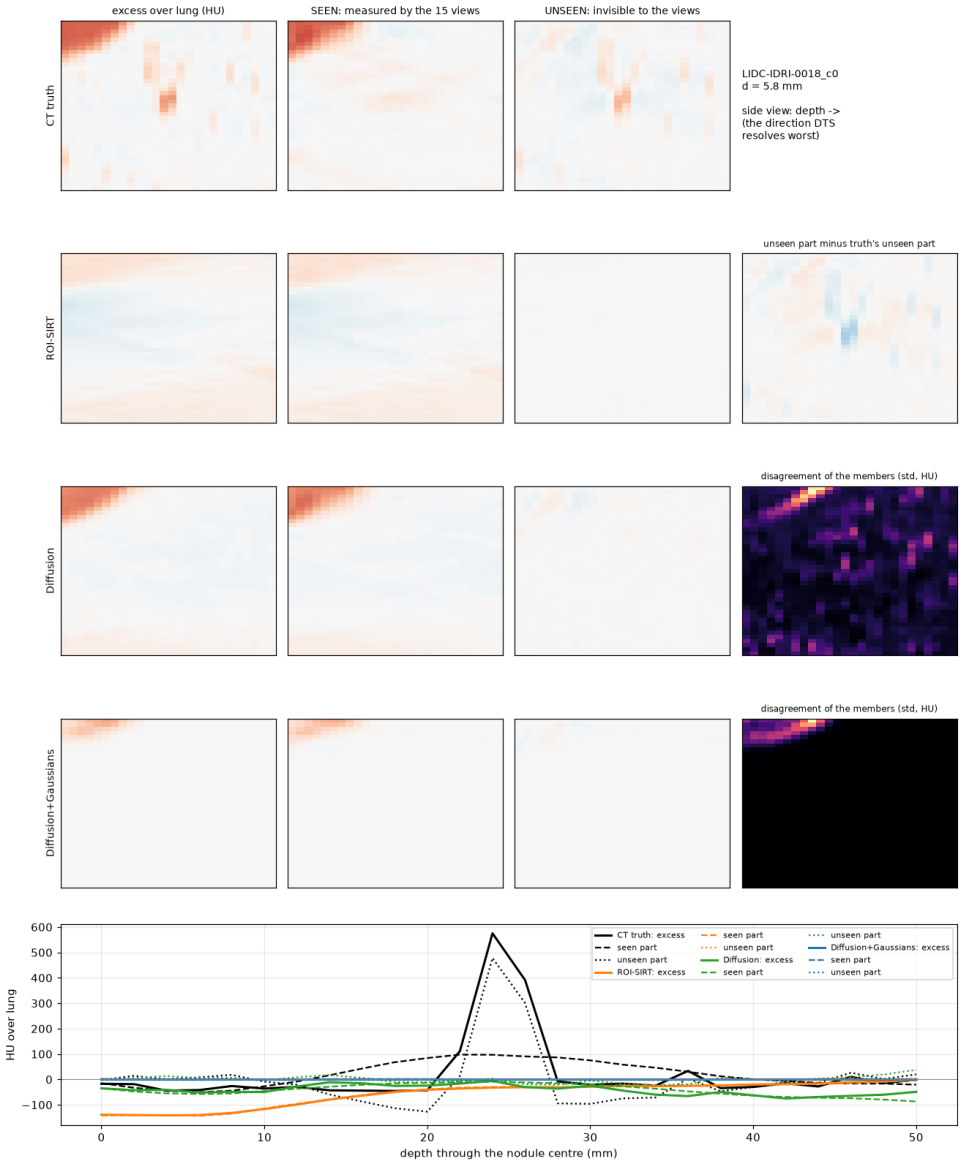


Figure 4.19: Seen/unseen split for one test nodule (LIDC-IDRI-0018): CT truth, ROI-SIRT, diffusion and diffusion with Gaussians, with the excess-attenuation profile through the nodule centre in depth (bottom).

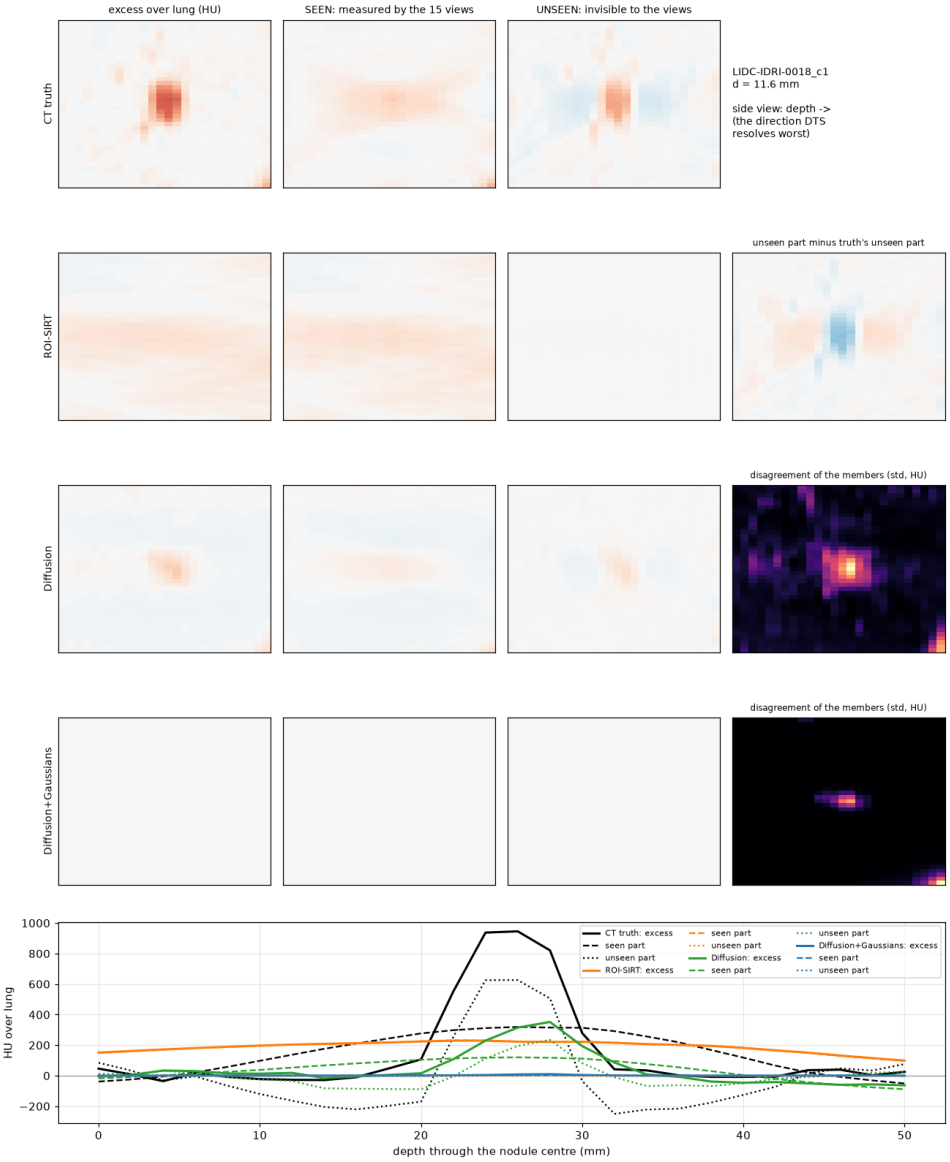


Figure 4.20: Seen/unseen split for a second test nodule (LIDC-IDRI-0018, 11.6 mm): the diffusion prior recovers depth that the views miss, while the Gaussian fit loses the nodule.

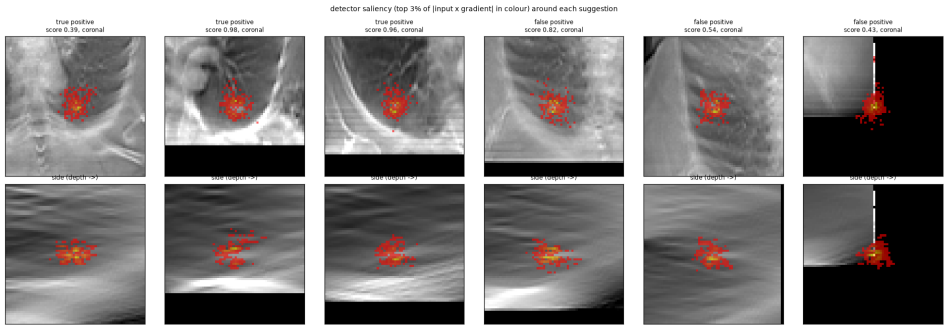


Figure 4.21: Input×gradient saliency of the detector for true and false positive suggestions (coronal and depth views).

4.3.7 Why the Measurements Failed

The results point to three causes, each of which suggests a fix.

1. **The prior shrinks faint nodules.** Training pairs were cut around nodules and around nodule-free lung, and the prior was not told that a nodule is present at the click. For faint nodules it often produces plausible normal lung instead, which is consistent with the strong negative bias, the larger errors of low-density nodules and the low ensemble spread on failures. Figure 4.22 shows such a case. A fix is to condition the prior on the click or oversample nodule pairs.

2. **Mass is measured in image space.** The X-rays fully define the mass (0.3% unseen), but our measurement is done by integrating the reconstruction in a segmented window, so there is a leakage from segmentation failures and from shrinkage of the prior, into the mass. As intended by the H1 physics, a projection-domain mass estimate would be an integration of the detector signal itself.

3. **Failures break the calibration.** The log-errors for measurements of zero are infinite and intervals are unbounded. Reporting "not measurable" as a different outcome with its own probability and calibrating only those measurable would result in finite intervals.

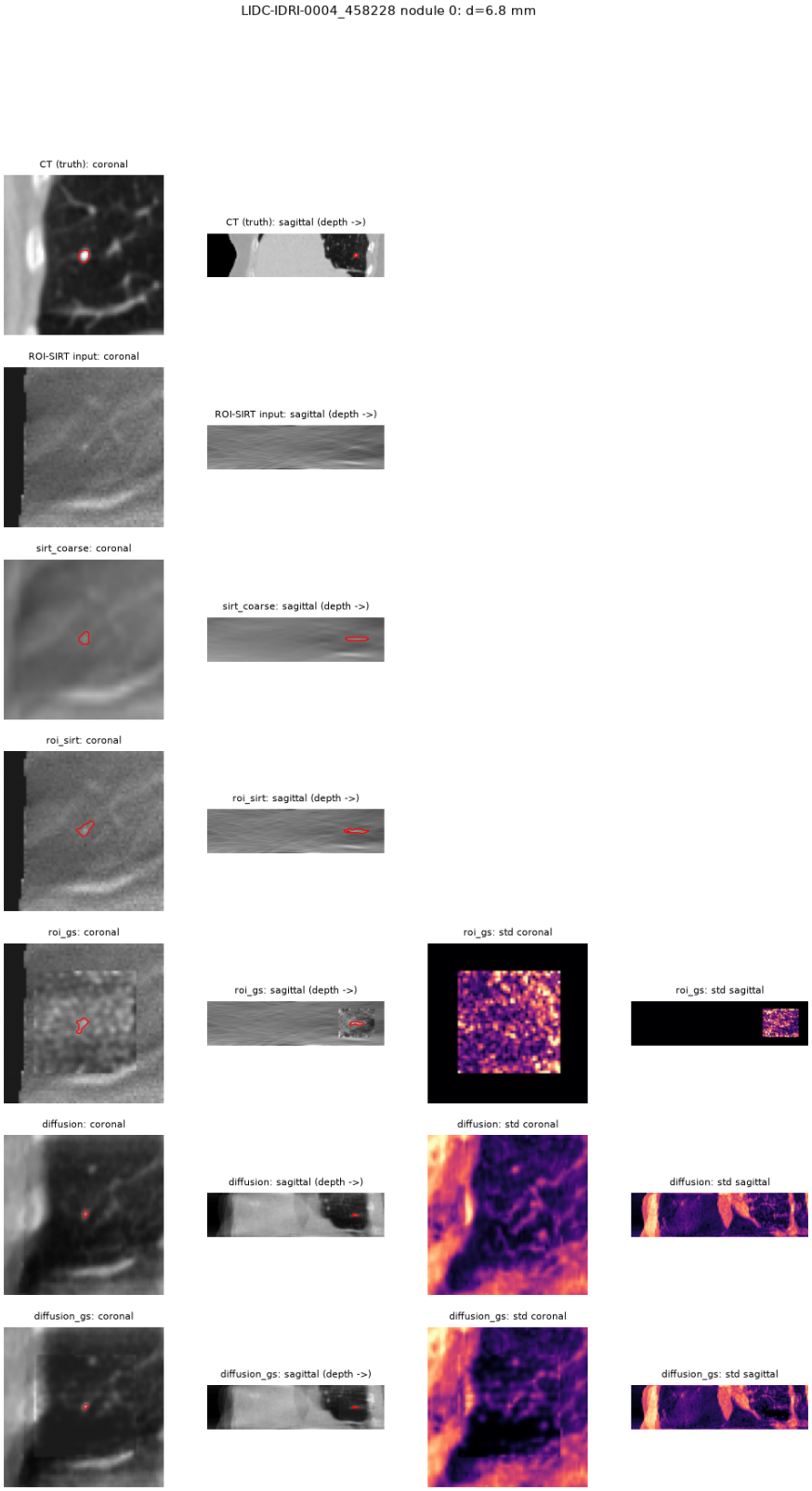


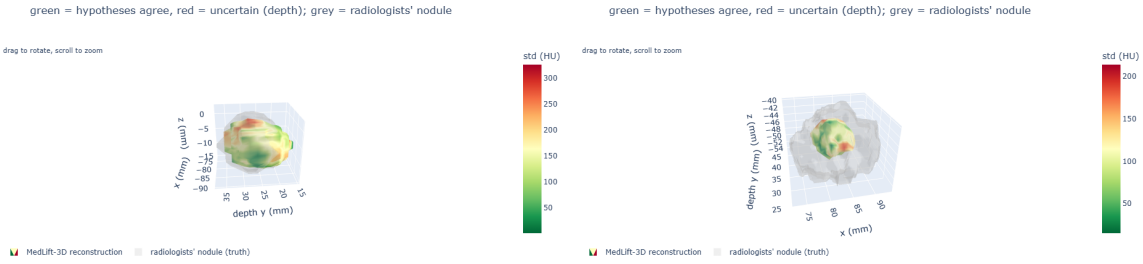
Figure 4.22: One test nodule (LIDC-IDRI-0004, 6.8 mm) through all methods: CT truth, the DTS input, and each reconstruction with its segmentation (red) and sample spread (right).

4.3.8 Acquisition Geometry Sensitivity

On 12 nodules of two test patients, the depth smear (depth over in-plane width of the artifact spread) was 5.13 for 15 views over 30°, 5.34 for 60 views over 30° (four times the compute), 4.69 for 15 views over 40° and 4.61 for 74 views over 40°, against 0.95 for CT. At equal dose the number of views hardly matters, while a 40° sweep reduces the smear by about 9%; our 15-view, 30° setting is therefore the conservative case.

4.3.9 Application

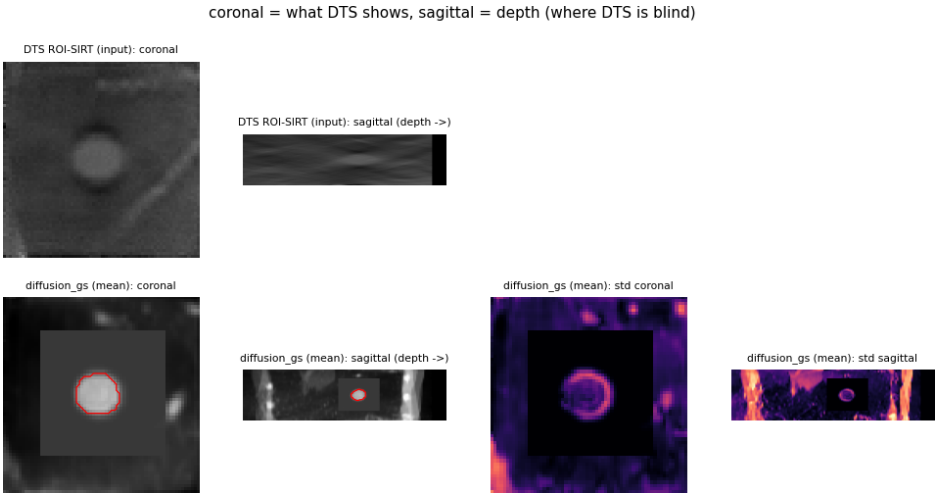
Measuring one nodule with all five reconstruction methods took a median of 45.5 s. The application loads the trained models on a free Kaggle notebook and is reached from a browser through an HTTPS tunnel. For a click on an annotated nodule it returns mass and volume with intervals, the EUPS category with probabilities, the prism views with the sample spread, and an interactive 3D view in which the reconstruction is coloured by the disagreement of the eight samples and the radiologists' nodule is shown in grey (Fig. 4.23); in both views the uncertainty is largest in depth, as the physics predicts. A click where no nodule exists returns "no nodule measured at the click" instead of a number, as NFR-05 requires.



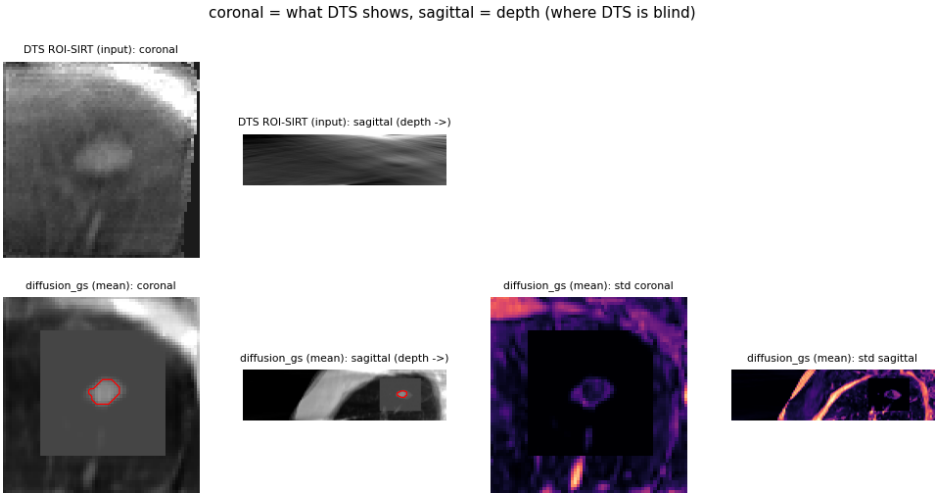
(a) **(b)**

Figure 4.23: 3D view of the application for two test nodules. The reconstruction is coloured by the disagreement of the eight samples (green: samples agree; red: uncertain, mostly in depth) and the radiologists' nodule is shown in grey. In (a) the reconstruction matches the consensus nodule; in (b) it is smaller than the consensus nodule, the shrinkage discussed in Section 4.3.

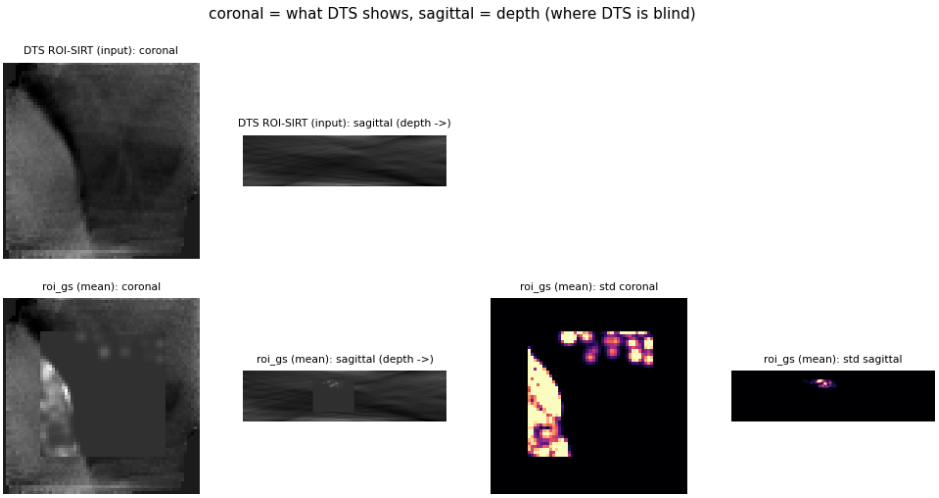
The application also shows the prism around the click in the coronal plane (what DTS shows) and the sagittal plane (depth, where DTS is blind), together with the mean reconstruction, its segmentation and the sample spread (Fig. 4.24). For a solid, well-contrasted nodule the segmentation follows the nodule and the spread is concentrated at its border; for a faint nodule the reconstruction is smaller and less certain; and the physics-only Gaussians can fail completely near the chest wall, where the spread is high everywhere and no nodule is found.



(a) Solid nodule, diffusion with Gaussians.



(b) Faint nodule, diffusion with Gaussians.



(c) Nodule near the chest wall, physics-only Gaussians: no nodule found.

Figure 4.24: Prism views in the application: DTS input (top row of each panel) and reconstruction with its segmentation (red) and sample standard deviation (bottom row), coronal and sagittal (depth).

4.4 Summary

All requirements that had a fixed criterion were automatically checked. As expected, the pipeline, the simulation, the reconstruction, the detector and the uncertainty machinery behaved as designed: the learned detector outperformed the training-free baseline by 17x in CPM, the diffusion prior improved nodule-region PSNR on the held-out pairs by 3.35 dB, conformal intervals achieved their target coverage, and the seen/unseen analysis demonstrated that the nodule's mass is completely captured by the X-rays and that the prior's "unseen" component correlates with the truth. None of the measurements in themselves achieved the objective – H1 and H2 were not achieved and the intervals, though valid, were too large to be of use – and our method did not achieve H4. These failures were attributed to the prior's shrinking of faint nodules, mass measurements in image space not from the projections, and failures in the measurement of nodules within the calibration set. These are not open-ended but concrete, testable fixes.

Chapter 5

Standards and Design Constraints

This chapter describes the standards that MedLift-3D adheres to (along with the alternatives considered), and examines the economic, environmental, ethical, health and safety, social, political, legal, cultural, and sustainability issues. It concludes by the cost analysis and complex engineering problem mapping.

5.1 Compliance with the Standards

MedLift-3D is research software to be developed as medical-device software, so we adhered to the standards which must be met by such software, although not formal certification for a student project. For every standard we explain how it is applied, the other options that were thought of with their pros and cons, and why the chosen one was selected. The Standards are divided into Software Standards, Hardware and Communication Standards.

5.1.1 Software Standards

The software standards are related to data formats, the software life cycle, risk and quality management, code quality and numerical precision and licensing (Table 5.1). The input path reads DICOM (the format in which hospitals store and exchange images) directly, and checks its geometry, while intermediate data are stored as compressed NumPy files with a JSON header, which store the volume, masks and metadata of each case in one self-describing file. We have not been certified to IEC 62304, but we did follow the core practices: each requirement was mapped to a test, all code is version controlled, 69 unit tests are run automatically and any anomaly discovered during software development has been documented (Chapter 4) by implementation of the life cycle processes defined in IEC 62304. The hazard analysis in Table 5.4 is used for the application of ISO 14971 risk management and ISO 13485 quality management is identified as required for a future product. This is intentional: physics, measurement is done in 32-bit precision; only diffusion network is done in bf16 to keep the 32-bit range, avoiding overflow of fp16 in 3D networks. Lastly, all data and libraries have open licence with attribution.

Table 5.1: Software standards.

Standard	How it is applied	Alternatives (pros / cons)	Rationale
DICOM [56]	Input CT read from DICOM; geometry checked against pylidc (0.000 mm difference)	NIfTI: simpler but loses scanner metadata	Hospitals and PACS store images in DICOM
Compressed NumPy with JSON headers	Cases, pairs and results keep their geometry in a header	NIfTI cannot hold per-nodule reader masks with metadata; HDF5 is a heavier dependency	One self-describing file per case
IEC 62304 [57] (software life cycle)	Not certified; practices followed: requirements traced to tests (Table 4.4), version control, unit tests, documented anomalies	Ad-hoc research code: faster but unverifiable	Needed for any clinical use; tracing now costs little
ISO 14971 [58] (risk management)	Hazard analysis with mitigations (Table 5.4)	No risk analysis	Required by regulators
ISO 13485 [59] (quality management)	Identified as required for a product; out of scope here	–	Applies to a device manufacturer
PEP 8, pytest	Code style; 69 automated tests	No tests: faster now, bugs later	Readable and checkable team code
IEEE 754	fp32 for physics and measurement; bf16 only inside the network	fp16 overflows in 3D U-Nets; fp64 doubles memory	bf16 keeps the fp32 range
Open licences	LIDC-IDRI under CC BY 3.0 with citation [60]; PyTorch (BSD), crepes (BSD-3), Gradio (Apache 2.0)	Proprietary data or libraries: costly, restrictive	Legal reuse with attribution

5.1.2 Hardware Standards

MedLift-3D is software, so the hardware standards concern the computing platform and the X-ray equipment that would produce real data (Table 5.2). All GPU work uses NVIDIA's CUDA platform through PyTorch, because CUDA GPUs are the ones offered by every cloud provider we could afford; AMD's ROCm platform is an open alternative but is rarely available for rent. The safety of the X-ray equipment itself is governed by IEC 60601-1 and its part 2-54 for radiography; these apply to the DTS unit, not to our software, and a clinical deployment would rely on a certified unit. Radiation doses are compared as effective dose following ICRP Publication 103, the standard measure of patient risk.

Table 5.2: Hardware standards.

Standard	How it is applied	Alternatives	Rationale
NVIDIA CUDA platform	All GPU computation through PyTorch's CUDA backend	AMD ROCm: fewer cloud offers; CPU only: training would take weeks	Widest availability on rented GPUs
IEC 60601-1 / 60601-2-54 (X-ray equipment safety)	Applies to the DTS unit that would produce real data, not to our software	–	We rely on certified DTS hardware
ICRP 103 effective dose [61]	Doses compared in mSv for CT, low-dose CT and DTS	Air kerma: less meaningful for patients	Standard measure of patient risk

5.1.3 Communication Standards

The communication standards concern how the application is reached and how results are exchanged (Table 5.3). The application runs as a web server and is reached over HTTP locally and over an encrypted HTTPS tunnel from outside, so any modern browser can use it without installing software; the usual Gradio share link failed on Kaggle, and SSH port forwarding would require setup on the client. Reports, results and calibration files are written as JSON, which is human-readable and language-neutral, unlike Python pickle files, which are unsafe to load from untrusted sources. Integration with hospital systems would use DICOM networking for images and HL7 FHIR for reports, the standards used by picture archiving and electronic record systems.

Table 5.3: Communication standards.

Standard	How it is applied	Alternatives	Rationale
HTTP/HTTPS with TLS	Application served locally over HTTP and to the browser through an HTTPS tunnel	Gradio share links (failed on Kaggle); SSH port forwarding (client setup)	Encrypted; works from any browser
JSON (RFC 8259)	Reports, results and calibration files	XML: verbose; pickle: unsafe to load	Human-readable, language-neutral
DICOM networking, HL7 FHIR	Future integration with hospital PACS and records	Custom APIs: no interoperability	Required in hospital workflows

5.2 Design Constraints

Economics, environmental, ethical, health and safety, social, political, sustainability, legal and cultural restrictions influenced the design. Each will be discussed below in conjunction with design impact and action taken.

5.2.1 Economic Constraint

There was no outside funding for the project, so all calculations had to be within a student budget. Wherever possible, free resources were used (laptop CPU, Kaggle GPUs for pilot and the application), CPU-only steps (download and case building) were delegated to a cheap CPU machine, and GPU training was allocated a fixed compute budget. This, in terms of clinical practice, boils down to economic factors: a DTS capable radiography room is significantly less expensive than a CT scanner, and can examine patients much faster, meaning that DTS can reduce the cost of follow-up for both hospitals and patients.

5.2.2 Environmental Constraint

Energy and emissions. The energy consumption of the L40S GPU was minimal, as was the emissions (assuming the per kWh emissions in the grid are approximately 0.48 kg CO_2 globally and approximately 0.6 in Bangladesh), for the entire study was conducted on a single GPU. About 0.004 kWh is consumed for 1 measurement (approx. 45 seconds on the GPU). These are the estimates based on rated power as per [62] and not metered values.

Data storage and waste. Downloaded data (64 GB of DICOM) was deleted after being converted to compact cases and intermediate files which could be regenerated were not backed up. No electronic waste was generated as no hardware was purchased.

Radiation as an environmental factor. This will decrease the collective dose of radiation for the population because CT follow-up (1.5-6 mSv) is replaced by DTS (0.1

mSv).

Mitigations. A small model (17.4 M parameters) instead of a large generator; a prism instead of whole-volume generation; fixed compute budgets; CPU work on CPU machines; trained models reused for evaluation and the application instead of retraining; and, for future runs, cloud regions powered by renewable energy.

5.2.3 Ethical Constraint

Professional ethics. We adhered to IEEE [63] and ACM [64] codes of ethics including honesty in claiming and results.

- **Honest claims:** all results are obtained from *simulated* DTS, not actual scanners, and Section 2.3 provides a list of the things we don't claim.
- **Honest results:** hypotheses were given prior to results; and the results in the report were whatever we stumbled upon such as: H1 and H2 were not supported; intervals were so wide as to be useless; detector sensitivity is only 35% at 2 false positives per scan; failed checks in Chapter 4.
- **Credit:** all datasets, libraries and ideas we build on are cited, including the data citations mandated by TCIA.
- **Fairness:** LIDC-IDRI is primarily derived from patients in the United States, and performance has not been determined on other patient populations such as patients with healed tuberculosis scars and is not claimed for such populations.
- **Data ethics:** Public data used under its licence, no re-identification attempt, Only de-identified data.

Professional responsibility. The team takes responsibility for their choices, and their errors. That we lost a third of the attenuation on the prism in the real data pilot, we found out and corrected ourselves on the lack of depth support in our design on the project before the full run; we also narrowed the first design rather than claim training losses as successes. The owner of each step (Table 3.7) is responsible for the checks for that step. The system is decision support – diagnosis remains the responsibility of the radiologist.

5.2.4 Health and Safety Constraint

Health impacts. *Physical:* Lower radiation for each follow-up (DTS ∼0.1 mSv, vs ∼1.5 mSv for low dose CT or several mSv for diagnostic CT) is important given nodule follow-up is repeated over many years [6, 15]. The most significant physical risk is mismeasurement, which can cause a cancer to grow and not be treated or can cause an unnecessary biopsy. *Mental:* having an indeterminate nodule is stressful, and a reassuring or alarming number aggravates it.

Actionable solutions. Each number is displayed with a calibrated interval and category probabilities to inform the clinician when a result is uncertain; a flag for low confidence triggers the clinician to consider CT and the results should be explained to the patient as categories instead of numbers. The principle hazards are listed in Table 5.4.

Table 5.4: Safety risk analysis (ISO 14971 style).

Hazard	Cause	Effect	Mitigation
Underestimated growth	Prior fills depth wrongly	Delayed treatment	Mass-first reporting; intervals; seen/unseen explanation; CT recommended when flagged
Overestimated growth	Vessel or chest wall included	Unnecessary CT or biopsy	Segmentation confined to a plausible extent; leaks reported as not measurable
Hallucinated nodule	Prior invents structure	False alarm	Data consistency with X-rays; clinician confirms
Missed nodule	Detector sensitivity 35% at 2 FP/scan	Nodule not suggested	Clinician can click anywhere; detector is an aid
Out-of-distribution use	Other population, scanner or dose	Silent errors	Validated setting stated; local validation and recalibration required
Over-trust	Automation bias	Clinician stops checking	Intervals and flags shown prominently; user training

5.2.5 Social Constraint

Short term, DTS-based follow-up would help to decrease CT waiting lists and patient travel outside of larger cities. It could enable nodule surveillance and ultimately screening in the longer term wherever CT resource availability is the limiting factor and the most survival benefit is achieved through early diagnosis. This translates to less dose, less cost for the patient, a measurement tool with honest uncertainty for the radiologist and local validation and training, and less cost for the health system for follow-up. Two potential adverse effects are the overusing of the tool, where many more people are followed, and the potential for a divide to develop between those who use the tool in private hospitals and those who don't, which is making their own guidelines; both suggest a need for public-sector implementation and adherence to guidelines.

5.2.6 Political Constraint

Public health policy and procurement are key enablers for health technology adoption. In Bangladesh, imaging equipment and software is mostly bought via the government tender process and large private hospital conglomerates, so the adoption of such equipment would require proof to be accepted by health authorities and incorporation into national cancer-control plans. The deployment of foreign cloud computing for health data is also a policy issue – data and computation should remain in the country.

5.2.7 Sustainability

Lifecycle plan. The various steps are: (1) development and verification (this project); (2) retrospective validation on real DTS with CT ground truth from a partner hospital; (3) a prospective reader study; (4) regulatory registration; (5) deployment with monitoring of measurement errors and interval coverage; and (6) periodic recalibration of the conformal layer, which only requires new calibration cases, not retraining, when scanners or populations change.

Three dimensions. *Environmental:* small model, low per measurement energy, no special hardware. *Economic:* uses one mid-range GPU or low cost cloud instance and is open-source software, no licence fees to be paid. *Social:* demonstrated and validated code for other team to maintain and process that can be extended to other limited angle modalities.

5.2.8 Legal Considerations

- **Data licence:** LIDC-IDRI is distributed by TCIA under the Creative Commons Attribution 3.0 Unported licence, which allows commercial, scientific and educational use with attribution. The TCIA data usage policy forbids any attempt to identify or contact the participants and requires every publication to cite both the collection [19] and the TCIA repository [65] [60]. We follow both requirements and made no attempt at re-identification.
- **Medical device regulation in Bangladesh:** medical devices are regulated by the Directorate General of Drug Administration (DGDA). The Drugs and Cosmetics Act 2023 [66], passed by Parliament in September 2023, replaced the Drugs Act 1940 and the Drugs (Control) Ordinance 1982 and includes provisions on medical devices. Registration follows the DGDA Registration Guidelines for Medical Devices [67], which are based on the Global Harmonization Task Force guidance and define a medical device to include software intended for diagnosis or monitoring. A clinical version of MedLift-3D would therefore be a registrable medical device; implementing rules under the 2023 Act are still being developed, so the current DGDA requirements must be confirmed at the time of submission.
- **Other markets:** US FDA (510(k) or De Novo), EU Medical Device Regulation [68] and EU AI Act [69] (high risk medical AI) would apply, all of which entail risk management [58], life-cycle processes [57] and clinical evidence.
- **Data protection:** The Personal Data Protection Ordinance 2025 [70] approved on 9 October 2025 and published in the official gazette on 6 November 2025 provides for citizens as owners of their personal data, specific consent is a requirement for collecting, storing, transferring, or using personal data, health data is classified as sensitive and subject to stricter rules and penalties, and cross-border transfers are

restricted; some provisions will come into effect 18 months after publication. For a clinical deployment, you'd need to get consent, comply with these regulations (and GDPR/HIPAA for data collected outside of the country) and ideally compute within the country. Only public data, which has been de-identified, is used in the demonstration.

- **Liability:** intended for decision support, diagnosis is the responsibility of the clinician.

5.2.9 Cultural Considerations

- **Language and literacy:** an interface to be deployed should support Bangla and output results in lay categories, numbers to be made available to specialists.
- **Family-centred decisions:** Diagnoses are often made with the family, so if there is any uncertainty this should be explained by the clinician not in the form of raw probabilities, but in a way the family can follow.
- **Stigma:** Lung cancer is associated with smoking and blame – do not blame and many patients, particularly women, have never smoked and are exposed to cooking smoke.
- **Local disease patterns:** Healed tuberculosis leaves behind calcified granulomas which look like nodules. Such patients were not included in the training of the model, and therefore, local validation is necessary before using the model, in accordance with the patients' right to a model that is tested on patients similar to themselves.
- **Trust:** many of the patients trust the doctor more than the machine and it is only fair to keep the clinician in charge.

5.3 Cost Analysis

Revenue model (illustrative). Assume 18 months for the commercial version of validation and registration and target customers to be diagnostic centres and hospitals that have DTS-capable radiography (Tables 5.5 and 5.6).

Table 5.5: Illustrative revenue (BDT lakh).

Revenue stream (price)	Year 1	Year 2	Year 3
Pay per study (BDT 300 per nodule follow-up)	0	18 (6,000 studies)	60 (20,000 studies)
Hospital subscription (BDT 6 lakh per year)	0	18 (3 hospitals)	48 (8 hospitals)
Total revenue	**0**	**36**	**108**

Under these assumptions the venture breaks even early in year four. The largest uncertainty is the cost and duration of clinical validation, which is why a partnership with a public hospital is the first step.

Table 5.6: Illustrative expenses and cash flow (BDT lakh).

Item	Year 1	Year 2	Year 3
Development team (3 engineers)	25	25	30
Clinical validation study	20	10	–
Regulatory registration and quality system	8	4	4
Cloud / GPU and hosting	3	5	8
Marketing and training	2	6	8
Total expenses	**58**	**50**	**50**
Net cash flow	−58	−14	+58
Cumulative cash flow	**−58**	**−72**	**−14**

5.4 Complex Engineering Problem

MedLift-3D is a complex engineering problem: it is ill-posed by physics, combines X-ray physics, generative AI and statistics, has conflicting requirements (dose, accuracy, honesty, cost), and has direct consequences for patients.

5.4.1 Complex Problem Solving

Table 5.7: Mapping with complex problem solving.

P1 Depth of knowledge	P2 Range of conflicting requirements	P3 Depth of analysis	P4 Familiarity of issues	P5 Extent of applicable codes	P6 Stakeholder involvement	P7 Inter-dependence
✓	✓	✓	✓	✓	◦	✓

P1 – Depth of knowledge. What is needed extra to the standard curriculum is X-ray projection physics, linear algebra of range and null spaces, probabilistic diffusion models, conformal statistics and 3D deep learning. The knowledge profiles are shown in Table 5.8.

Table 5.8: Knowledge profile mapping for P1.

Knowledge profile	Where it is used
WK1 Natural sciences	X-ray attenuation, Poisson photon statistics, radiation dose
WK2 Mathematics	Projection moments and conditioning; CGLS; probability and exchangeability for conformal prediction
WK3 Engineering fundamentals	Numerical methods (SIRT, conjugate gradients), algorithms, data structures
WK4 Specialist knowledge	Tomography, diffusion models, 3D U-Nets, Gaussian representations
WK5 Engineering design	Requirements, prism design, verification steps, application
WK6 Engineering practice	Cloud GPU operation, testing, version control, reproducibility
WK7 Engineering in society	Radiation safety, regulation, ethics, cultural factors
WK8 Research literature	Literature on DTS, sparse-view CT, diffusion, Gaussians and conformal prediction

P2 – Range of conflicting requirements. The dose vs accuracy, the filling of the missing wedge vs no anatomical invention, the single numbers vs honest intervals, the memory vs the field of view, and the detection recall vs false positives (Table 3.2).

P3 – Depth of analysis. We started from the ground up to identify what DTS can measure, created a virtual 583 patient trial and looked at calibration, size group coverage and seen/unseen content not just image quality.

P4 – Familiarity of issues. For the calibrated 3D nodule measurement in DTS, no standard solution exists and already the combination of a data-consistent prior in a DTS-prism, Gaussian mass and conformal calibration had to be designed.

P5 – Extent of applicable codes. DICOM, IEC 62304, ISO 14971, data licences and medical device regulation (Table 5.1)

P6 – Extent of stakeholder involvement (partial). Clinical needs were based on guidelines and feedback from supervisor; future work will include direct evaluation by the radiologists.

P7 – Interdependence. There is a lot of interdependence between simulation, reconstruction, prior, Gaussians, calibration, and detector: for instance, the depth-support bug in reconstruction screwed up the training data, and consequently all subsequent measurements, including calibration and detector, until it was fixed.

5.4.2 Engineering Activities

Table 5.9: Mapping with complex engineering activities.

A1 Range of resources	A2 Level of interaction	A3 Innovation	A4 Consequences for society and environment	A5 Familiarity
✓	✓	✓	✓	✓

A1 – Range of resources. There was a public collection of 583 patients (≈64 GB), rented CPU and GPU machines with shared storage, free Kaggle GPUs and open-source libraries.

A2 – Level of interaction. Three persons in the team (with different roles), the supervisor, and the course teacher, published clinical guidelines, and open-source communities (crepes, Gradio, pylidc) interact.

A3 – Innovation. The mass-first projection-moment argument for DTS, the prism-based data-consistent diffusion prior, decision-level calibrated uncertainty and a seen/unseen test with respect to the truth.

A4 – Consequences for society and environment. The intervals, flags and clinician control all address the potential for incorrect measurements, while providing lower radiation and cost per follow-up and greater access to nodule surveillance (Table 5.4).

A5 – Familiarity. During the project, diffusion models, conformal predictive systems and null-space analysis were learned, which are much more than typical undergraduate projects.

5.5 Summary

MedLift-3D follows the software, hardware and communication standards relevant to medical imaging software and prepares for IEC 62304 and ISO 14971. Its main impacts are positive (lower dose and cost, wider access), with clear risks (wrong measurements, overtrust, untested populations) that the design mitigates through calibrated intervals, flags, clinician control and a validation plan. It meets the attributes of a complex engineering problem.

Chapter 6

Conclusion

The end of this chapter summarises the work completed by MedLift-3D, its limitations and future work, and introspects on the independent learning needed by the project.

6.1 Summary

MedLift-3D converts a single click on a low dose chest tomosynthesis image to a 3D measurement of the nodule, and provides the calibrated uncertainty measurement. We demonstrated via projection moments that DTS measures a nodule's mass in all views but its depth extent about 15 times worse, and built the system around this fact: a prism reconstruction around the click, a 17.4-million-parameter conditional diffusion prior that was consistent with the X-rays, 3D Gaussians that give mass in closed form, conformal intervals and calibrated size-category probabilities, a learned detector to suggest clicks, and a seen/unseen explanation of every result. A virtual clinical trial was conducted on LIDC-IDRI, and verified 18-step pipeline ran the trial with 583 patients. The components performed as desired: the prior raised the nodule-region PSNR from 11.88 dB to 15.23 dB on the held-out data, the detector achieved a CPM of 0.298, versus 0.017 for a training-free baseline, the conformal intervals had their desired coverage, the calibrated probabilities reduced the Brier score for $P(V \geq 100)$ from 0.54 to 0.17, and the null-space analysis demonstrated that the X-rays measure nearly all of a nodule's mass and that the prior's unseen content correlates with the truth. The measurements did not meet the desired accuracy: Mass was not more accurate than Volume (H1); the whole method was not better than the physics-only reconstruction method (H2); MDT was not better than VDT with our method (H4); and the valid intervals were too large to be useful (H3). Calibrated probabilities (H5) were helpful at one of two thresholds, while the positive (unseen) prior skill from the model favors H6. The analysis revealed the failures to specific causes and fixes and demonstrated that it was not the case of better image quality is better measurement – also a lesson for medical generative AI.

6.2 Limitation

- **Simulation only.** There is no public chest DTS dataset with CT ground truth; the performance of the DTS on real scanners is unknown since our DTS is simulated using CT, monochromatic beam and no scatter.
- **Acquisition.** For example 15 views as opposed to the 60-74 views of clinical units (at equal dose) and a conservative 30° sweep; the prior was trained and tested only at this geometry.
- **Detector sensitivity.** 35% at 2 false positives per scan is not sufficient for unaided use - detector is an aid only.
- **Growth.** The growth of the virtual nodule is uniform whereas real nodules grow irregularly.
- **Ground truth.** LIDC labels vary between radiologists (consensus volume / mean reader volume 0.81), and only nodules with a diameter of 4–30 mm which were marked by at least 3 readers were used.
- **Population.** LIDC comprised primarily of US patients, and did not represent populations with tuberculosis scarring.
- **Measurement accuracy.** Volume and mass errors (MdAPE 75–100%) are unsuitable for clinical practice and the whole method under-estimates considerably (bias −91%) and fails in 22% of nodules.
- **Uninformative intervals.** The intervals were valid and unbounded, and the ensemble spread was negatively correlated with the error due to failed measurements in the calibration set.
- **Training budget and size.** The prior continued to improve at the end of training, and its samples were more useful for training patients than for new ones (+7.00 vs +4.22 dB), and the detector also partially memorised its training patients.
- **Sample size.** The numbers of nodules used for measurements were 150 (76 calibration and 74 test) and 40 growth pairs and the difference between the methods is small and therefore not conclusive.
- **Patient thickness** was obtained from the CT body outline and for a real DTS unit it would have to be measured.

6.3 Future Work

1. **Nodule-preserving prior:** condition the diffusion model on the click position, and oversample the pairs of nodules, so that faint nodules aren't replaced by normal lung.

2. **Projection-domain mass:** use integration of the detector signal around the nodule's shadow, which is the null-space analysis' indicator of the signal being fully measured, as opposed to integration of a segmented reconstruction.
3. **Failure-aware calibration:** Intervals are calibrated on measurable cases only, and additional outcome is "not measurable" with its own probability and a difficulty score is positively correlated with error.
4. **Real DTS data:** collab with a hospital with a DTS unit to obtain paired DTS and CT scans, re-calibrate the conformal layer on these and measure the real performance.
5. **Physics realism:** incorporate polychromatic spectra, scatter and the clinical protocols (60–74-view) in the simulator.
6. **Longer and larger training:** train the prior longer with augmentation and test the larger models and train detector with more nodule-free negatives.
7. **Real growth:** don't use virtual growth, instead use longitudinal CT pairs.
8. **Reader study:** evaluate the accuracy of follow-up decision making, when using MedLift-3D in addition to DTS, compared to DTS-only use.
9. **Deployment:** a Bangla interface, DICOM/PACS integration, an on-site GPU option and a quality system towards registration.
10. **Other limited-angle problems:** breast tomosynthesis and dental cone-beam CT which have the commonality of the missing-wedge structure.

6.4 Independent Learning and Reflection

This project involved much more knowledge and skill than was being taught in our course. In this section, we look back at that independent learning under five aspects: curiosity, self-initiative, independence, transfer of past learning, reflection on what we learned.

6.4.1 Curiosity

This was originally conceived as a generative-modelling concept but we were continually asking ourselves "what can a 30° X-ray sweep actually measure?" Taking that question over one of the projection moments resulted in the mass-first design. Due to this curiosity, we wondered if the prior recovered or invented depth, and, in fact, the seen/unseen analysis came into existence and we were able to produce calibrated category probabilities only after studying conformal prediction.

6.4.2 Self-initiative

In addition to the mandatory scope we included a learned detector after the training-free detector, which mainly detected vessels, a fit check to demonstrate that no overfitting was occurring in the prior, conformal predictive systems with size-group calibration, explainability with a truth referenced test and a geometry sensitivity study to justify the 15-view choice, and an automatic HTML report that made all checks visible to examiners.

6.4.3 Independence

In addition to the classroom, we were able to learn about running a long unattended job on rented cloud machines, downloading and organizing 64 GB of medical images from TCIA, using pylidc to read radiologist annotations and serving a GPU application from a Kaggle notebook via an HTTPS tunnel when the traditional share link didn't work.

6.4.4 Transfer of Past Learning

A lot of the project is based on knowledge developed in previous courses which is applied to a problem none of the courses dealt with. Linear algebra was used to disentangle what the X-rays measured from what the prior supplies; calculus was used to give the moment argument that underpinned the design; probability and statistics became calibrated uncertainty; and software engineering was used to turn the design into a verified pipeline. Table 6.1 shows the areas in which each application was made.

Table 6.1: Earlier coursework applied in MedLift-3D.

Earlier course knowledge	Application in MedLift-3D
Linear algebra (range, null space, least squares)	Seen/unseen split with CGLS; SIRT as an iterative least-squares solver
Calculus and moments	Projection-moment argument: mass is the 0th moment, depth extent the 2nd
Probability and statistics	Poisson noise, conformal prediction, Brier score, rank correlations, bootstrap confidence intervals
Numerical methods	Iterative solvers, numerical stability, mixed precision
Data structures and algorithms	Ray marching, candidate ranking, patient-level splitting
Artificial intelligence and machine learning	Diffusion training, U-Nets, overfitting checks, train/calibration/test discipline
Image processing and computer graphics	Projection geometry, 3D Gaussians, segmentation, saliency maps
Software engineering	Requirements and traceability, unit tests, version control, the verified pipeline
Computer networks	HTTPS tunnels and port forwarding for the application

6.4.5 Reflection on Learning

Our biggest lesson is that a large architecture with a decreasing training loss does not mean that one clinical measurement is correct. The restructuring of the work on measurable

clinical numbers, explicit hypotheses and automatic checks had an impact on the way people think about engineering: if we can't verify a result, it isn't a result; if it is a negative result, it is more valuable if we tell the truth. We also learned to work within real constraints of time, money and data, to budget our computation as well as to embrace uncertainty as part of our answer instead of an embarrassment. It's not so much a model we take away from the project as these habits.

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